Arithmetic of Gieseking Moment Lattices: Normality, Integral Saturation, and Approximation Budgets

Keio

21 October 2026

Abstract

We study the four formal period coordinates generated by the kernel $(1 - s^2t^2/4)^{-1/2}$ and the Gieseking volume $V = \mathrm{Cl}_2(\pi/3)$. A synthesis of earlier and subsequent moment constructions leads to several further results. Three powers of a short least common multiple suffice for the target columns of the complete moment inverse, settling the exponent-three question left in the supplied inverse construction. We classify the entire first residue layer of coefficient saturation in the no-wrap regime and close a previously undetermined two-block prime interval. For a direct odd output space we determine the exact dyadic integral-output index and obtain an exponential lower bound for every complete integral output basis. We optimize the positive Bernstein approximation budget, control exceptional primes for growing conic normalizers, and identify the irrationality exponent encoded by a calibrated moment lattice. Finally, prime-scale coefficient certificates exclude algebraic roots of height below $2\log p$, giving a degree-explicit criterion for future transcendence arguments. Complete finite-window normality, local image calculations, and the inverse construction are included as foundations. These results resolve specific structural and arithmetic questions within the constructions. They do not prove irrationality or transcendence of V: the needed infinite smallness estimate for genuinely integral surviving forms remains unproved.

Keywords. Gieseking volume; Clausen function; period coordinates; Legendre polynomials; local saturation; denominator bounds; Diophantine approximation.

Contents

1 Introduction and the new conclusions 3

2 Canonical moments and conventions 4

3 Exact finite-window normality 5

4 Constructing the complete moment inverse 10

5 Three short-lcm powers suffice for the target inverse columns 14

6 Sharp local period images and denominator thresholds 16

7 Optimizing the Bernstein degree and denominator budget 21

8 Complete first-layer saturation and the missing two-block interval 25

9 Quantitative arithmetic of the conic normalizers 29

10 An exact dyadic index and an obstruction to short integral output bases 32

11 A calibrated one-row moment lattice 36

12 The exact Diophantine exponent of the calibrated moment lattice 39

13 Algebraic root exclusion and the missing transcendence inequality 41

14 What is closed, and what remains 43

A Source coverage and reproducibility 44

1 Introduction and the new conclusions

The constant under consideration is

$$V = \mathrm{Cl}_2(\pi/3) = \frac{3\sqrt{3}}{4} L(2, \chi_{-3}) = \int_0^1 \int_0^1 \frac{ds\, dt}{\sqrt{1 - s^2 t^2/4}} = 1.0149416064\ldots. \tag{1.1}$$

The arithmetic of an explicit period construction has several separate parts: a rational-coordinate map must be nondegenerate, its local images must be known, its denominators must be controlled, and a nonzero integral form must be small at the real place. Solving one of these parts does not automatically solve the others. This paper treats their interaction in a single moment system.

The source corpus consists of 136 earlier Gieseking notes and 47 subsequent notes, in five named series. We use the strongest compatible results and prove further consequences, rather than concatenate all 183 notes. The source identifiers in the bibliography refer to supplied, unpublished manuscripts. The word "new" below means a deduction made in this synthesis relative to those supplied statements; it is not a literature-wide priority claim. The finite checks are reproducible and supplement the proofs.

1.1 Results established by combining the constructions

The main advances are as follows.

1. **Target inverse denominator.** For every n and prime $p \geq 5$, each target column of the full $4n$-column inverse has valuation at least $-3\lfloor \log_p(2n-1) \rfloor$ (Theorem 5.1). The previously proved factor four is reduced to three. This resolves the explicit unproved improvement in Meiji R62; optimality of three is not claimed.

2. **Complete first-layer saturation.** Theorem 8.2 classifies all sources in a no-wrap window, not just selected divided columns. Only the two boundary prime blocks can enlarge its source span. Combining the resulting upper inclusion with the earlier nested-image lower inclusion closes $19/100 < p/n < 39/200$:

$$\Gamma_{\mathrm{sat},(\mathrm{p})} = \Gamma_{\mathrm{nested},(\mathrm{p})} = p^{-1}\mathbb{Z}_{(p)} \times p\mathbb{Z}_{(p)}.$$

   The polynomial exceptions and compact-interval qualification are retained in Theorem 8.4.

3. **An intrinsic integral-basis cost.** For the direct odd rational-output basis the exact dyadic index is

$$2^{4n^2 - 2n - 3\sum_{j<n} s_2(j)}.$$

   Together with the odd-prime residue ranks, this forces a geometric-mean coefficient norm of at least $\exp((4\log 2 + 1/12 + o(1))n)$ for every complete integral output basis (Theorem 10.2 and Corollary 10.3). This is a restriction on the specified rational-output space.

4. **Optimized positive approximation budget.** At full Bernstein width the best binomial order for the stated universal denominator is $m/n = 1/3$, and the sufficient control-height budget improves from $0.9175241689\ldots$ to $1.1755429311\ldots$ (Theorem 7.1). The necessary infinite sequence of controls is not constructed here.

5. **Uniform normalizer arithmetic.** Nonzero high-index normalizers have height $\exp(O(h))$; the product of their exceptional numerator and denominator primes through index H has logarithm $O(H^2)$ (Theorem 9.4).

6. **The precise meaning of a calibrated lattice.** Conditional on V being irrational, its irrationality exponent satisfies

$$\limsup \frac{-\log \lambda_1(2a)}{\log \binom{2a}{a}} = 1 - \frac{1}{\mu(V)}, \qquad \liminf \frac{-\log \lambda_2(2a)}{\log \binom{2a}{a}} = \frac{1}{\mu(V)}.$$

The rational alternative is also determined (Theorem 12.1).

7. **Algebraic specialization.** The exact prime-scale coefficient profiles exclude every algebraic root of height below $2\log p$, uniformly in its degree (Theorem 13.1). A resultant inequality then separates the extra decay needed for irrationality, degree exclusion, and transcendence (Theorem 13.4).

1.2 External arithmetic and the remaining target

Calegari, Dimitrov and Tang proved the $\mathbb{Q}$-linear independence of $1, \zeta(2), L(2, \chi_{-3})$ [1]; their later work gives effective irrationality measures for the last value [2]. The scalar $3\sqrt{3}/4$ in (1.1) is essential: an irrational number multiplied by a nonzero algebraic irrational number can be rational. In particular $V \in \mathbb{Q}$ would make $L(2, \chi_{-3})$ a rational multiple of $\sqrt{3}$, which is compatible with the quoted theorems. No extension of their scalar field is presumed here.

The positive-control criterion below would prove $V \notin \mathbb{Q}$ if its infinite height hypothesis were met. The determinant criterion would exclude algebraic degrees if its primitive real-place decay were met. Neither hypothesis is proved in this paper. The unconditional results are the explicit denominator, image, index, optimization, height, and calibration theorems listed above.

2 Canonical moments and conventions

For rational polynomials q, P, put

$$\mathcal{B}(q, P) = \int_0^1 \int_0^1 \frac{q(s)P(t)}{\sqrt{1 - s^2t^2/4}}\, ds\, dt.$$

Throughout the mixed-moment sections q is even. We use formal coordinates

$$\mathcal{B}(q, P) = \beta + \tau\sqrt{3} + \sigma\pi + \alpha V, \qquad (\beta, \tau, \sigma, \alpha) \in \mathbb{Q}^4,$$

defined by the recurrences, rather than by an assumption of rational independence of the four real constants. In the conic sections a different ordering is displayed explicitly. Thus a nonzero formal vector need not have nonzero real evaluation.

Let $J_j = \int_0^1 t^j (1 - t^2/4)^{-1/2}\, dt$ and $I_j = \int_0^1 -t^j \log t (1 - t^2/4)^{-1/2}\, dt$. Direct integration by parts gives

$$J_0 = \pi/3, \qquad J_1 = 4 - 2\sqrt{3}, \qquad I_0 = V,$$
$$J_j = \frac{4(j-1)}{j} J_{j-2} - \frac{2}{j}\sqrt{3},$$
$$I_j = \frac{4(j-1)}{j} I_{j-2} - \frac{4}{j} J_{j-2} + \frac{1}{j} J_j \quad (j \geq 2 \text{ even}).$$

For even i the mixed moment is

$$M_{ij} = \begin{cases} (J_j - J_i)/(i - j), & i \neq j, \\ I_i, & i = j. \end{cases}$$

Its target coordinate is zero off the diagonal and equals $\binom{i}{i/2}$ on the even diagonal. These identities specify all formal coordinates exactly. The integral identity for V follows either by integrating in s and substituting $t = 2\sin(u/2)$, or by the absolutely convergent positive series

$$V = \sum_{r\ge 0} \frac{\binom{2r}{r}}{16^r(2r+1)^2}.$$

The p-adic valuation is normalized by $v_p(p) = 1$ and $v_p(0) = +\infty$. The ring $\mathbb{Z}_{(p)}$ is the localization of $\mathbb{Z}$ at p; completing to $\mathbb{Z}_p$ does not change the finite local indices used below. A coefficient saturation is the intersection of a rational polynomial space with the integral coefficient lattice. All logarithms are natural. A primitive integer polynomial has coefficient gcd one. A sufficient denominator bound is distinguished from a least denominator.

3 Exact finite-window normality

3.1 The determinant and its provenance

The selected target is irrationality of $V = \mathrm{Cl}_2(\pi/3)$, the Gieseking volume. No such conclusion is claimed here. We use Taisho's formal moment algebra [5]. The three-block determinant of Heisei round06 [16] motivates the present extension; its branch-point multiplicity and infinity row-reduction arguments are adapted explicitly below. The new fourth block is the second endpoint sequence $d_j(0)$. Its presence changes the zero multiplicity at the origin, the factorial constant, and the resulting full-coordinate rank theorem. Heisei's later odd-column sign-regularity construction [18] is distinct and is not used.

Define

$$c_{2r} = \binom{2r}{r}, \qquad c_{2r+1} = 0,$$

$$d_0(x) = 0, \qquad d_1(x) = -2, \qquad d_j(x) = \frac{4(j-1)}{j} d_{j-2}(x) - \frac{2}{j} x^{j-1} \quad (j \ge 2).$$

For $n \ge 1$, let $\mathcal{H}_n(x)$ be the $4n$ by $4n$ matrix with columns $0 \le j < 4n$ and four successive row blocks

$$(j^k c_j)_j, \quad (j^k d_j(0))_j, \quad (j^k d_j(x))_j, \quad (j^k x^j)_j, \qquad 0 \le k < n. \tag{3.1}$$

Here $j^0 = 1$ even at $j = 0$.

Theorem 3.1. *For every $n \ge 1$,*

$$\det \mathcal{H}_n(x) = C_n x^{n(3n-1)} (x^2 - 4)^{n^2}, \tag{3.2}$$

where the positive rational constant is

$$C_n = 2^{4n^2-2n} \left(\prod_{i=0}^{n-1} \frac{(i!)^2((2i)!)^2}{(2n+2i+1)!} \right) \prod_{j=1}^{2n-1} j!. \tag{3.3}$$

Thus $\mathcal{H}_n(x)$ is nonsingular exactly when $x \notin \{0, 2, -2\}$.

We prove the formula by three multiplicities, a degree bound, and an explicitly nonzero leading coefficient. All local series manipulations are formal over $\mathbb{Q}$.

3.2 The two branch points and infinity

Induction in the recurrence gives

$$(4-x^2)d_j'(x) - xd_j(x) = 2(x^j - c_j), \qquad d_{2r+1}(0) = -\frac{2\,16^r(r!)^2}{(2r+1)!}. \tag{3.4}$$

Fix $b \in \{2,-2\}$ and write $z = x - b$. There is a unique formal series $a(x)$ at b satisfying $(4-x^2)a' - xa = -2$: the coefficient determining $[z^r]a$ is $-b(2r+1) \neq 0$. Subtract a times the first block from the third and divide column j by x^j, a unit locally. Put $F_j = x^{-j}(d_j - ac_j)$. Then

$$(4-x^2)F_j' + \big(j(4-x^2)/x - x\big)\,F_j = 2.$$

The coefficient of z^r in F_j is a polynomial in j of degree at most r, by induction; every occurrence of j in its recursion is multiplied by a series vanishing at b. The fourth block is now $v_k = (j^k)_j$, $k < n$. After quotienting by these rows, a nonzero determinant term from the third block needs n distinct degrees at least n. Its total order is therefore at least

$$\sum_{s=n}^{2n-1} s - \sum_{k=0}^{n-1} k = n^2.$$

The other two blocks do not affect this vanishing argument. Hence both $(x-2)^{n^2}$ and $(x+2)^{n^2}$ divide the determinant.

For the degree bound, divide column j by x^j at infinity. In the first and third blocks replace j^k by $P_k(j) = \prod_{s=0}^{k-1}(j-2s)$; in the second use $Q_k(j) = \prod_{s=0}^{k-1}(j-2s-1)$. These monic triangular basis changes have determinant one. The first and second block rows then have orders $O(x^{-2k})$ and $O(x^{-2k-1})$ respectively, by their parity and the displayed roots.

As in [16], the third row can be reduced to order $O(x^{-2k-1})$ by subtracting multiples of first- and fourth-block rows. Here are the details needed to avoid a degree assumption. Iteration of the recurrence gives

$$[x^{j-2r-1}]d_j(x) = a_r(j) := -2\,4^r\,\frac{\prod_{s=0}^{r-1}(j-2s-1)}{\prod_{s=0}^{r}(j-2s)} \quad (j \geq 2r+1).$$

For $r < k$, $P_k(z)a_r(z)$ is a polynomial of degree $k-1$. Subtract its row, multiplied by x^{-2r-1}, using the fourth block. The only exceptional indices are even $j = 0, 2, \ldots, 2k-2$, where the rational expression originally had a removable product of a zero and a pole. The resulting exceptional values are removed exactly with the first k rows of the first block: their matrix on those columns is triangular with diagonal $P_s(2s)c_{2s}x^{-2s} \neq 0$. Successive solution uses multipliers $O(x^{-1})$. On all other even columns these corrections are $O(x^{-2k-1})$, and on odd columns they vanish. This proves the claimed row order.

The divided determinant has order at most $x^{-\sum_{k<n}(2k+2k+1+2k+1)} = x^{-3n^2+n}$. Restoring the columns gives

$$\deg\det \mathcal{H}_n \leq \sum_{j=0}^{4n-1} j - 3n^2 + n = 5n^2 - n. \tag{3.5}$$

3.3 The origin and the factorial constant

This is the additional step specific to the four-block problem. Solving (3.4) formally at zero gives

$$d_j(x) = b(x)d_j(0) + a(x)c_j + b(x)\int_0^x \frac{t^j}{\sqrt{4-t^2}}\,dt,$$

where

$$b(x)=\frac{2}{\sqrt{4-x^2}},\qquad a(x)=-\frac{2}{\sqrt{4-x^2}}\int_0^x\frac{dt}{\sqrt{4-t^2}}.$$

Subtract the first two terms from the third block. Put $w_r=\binom{2r}{r}/16^r$. The remaining entry has expansion

$$x^{j+1}\sum_{r\ge0}R_r(j)x^{2r},\qquad R_r(z)=\frac12\sum_{v=0}^{r}\frac{w_{r-v}w_v}{z+2v+1}. \tag{3.6}$$

Replace the kth polynomial row factor in this block by $S_k(z)=\prod_{v=0}^{k-1}(z+2v+1)$. For $r<k$, S_kR_r is a polynomial of degree at most $k-1$, so its term can be subtracted using the fourth block. At $r=k$, subtract the polynomial part as well. The leading remaining row is

$$x^{j+2k+1}\frac{\lambda_k}{j+2k+1},\qquad \lambda_k=\tfrac12 w_kS_k(-2k-1)=\frac{(-1)^k(2k)!}{2\,8^k k!}. \tag{3.7}$$

All later terms have at least two additional powers of x.

Expand along the last $2n$ rows. For their column set J, their common factors contribute at least

$$\sum_{j\in J}j+\sum_{k=0}^{n-1}(2k+1)\ge n(2n-1)+n^2=n(3n-1).$$

Equality forces $J=\{0,\dots,2n-1\}$. The leading determinant on these columns is $\prod_{k<n}\lambda_k$ times

$$\mathcal{C}_n=\det\begin{pmatrix}((j+2k+1)^{-1})_{0\le k<n,\,0\le j<2n}\\ (j^k)_{0\le k<n,\,0\le j<2n}\end{pmatrix}=\frac{\mathcal{V}(0,\dots,2n-1)\mathcal{V}(1,3,\dots,2n-1)}{\prod_{j=0}^{2n-1}\prod_{k=0}^{n-1}(j+2k+1)}>0.$$

Here $\mathcal{V}$ is the increasing-node Vandermonde product. The identity follows by multiplying the columns by $\prod_{k<n}(j+2k+1)$ and comparing the resulting polynomial evaluation determinant. Alternatively, nonvanishing alone follows because a rational function $P(z)+\sum_{k<n}a_k/(z+2k+1)$ with $\deg P<n$ cannot have $2n$ distinct nonnegative zeros unless it is zero.

The remaining columns split into $E=(2n,2n+2,\dots,4n-2)$ and $O=(2n+1,2n+3,\dots,4n-1)$. Their first two block minors contribute $\mathcal{V}(E)\mathcal{V}(O)\prod_{j\in E}c_j\prod_{j\in O}d_j(0)$, up to the interlacing sign $(-1)^{n(n-1)/2}$. The sign from $\prod\lambda_k$ is the same, so the leading coefficient at zero has sign $(-1)^n$ and is nonzero. Dividing it by $(-4)^{n^2}$ yields a positive C_n. Substitution of (3.4), (3.7) and the Vandermonde products simplifies exactly to (3.3).

We have found coprime factors with total degree $n(3n-1)+2n^2=5n^2-n$, equal to (3.5), and a nonzero coefficient at that order. This proves Theorem 3.1.

3.4 Minimal-window formal moment normality

For clarity, the actual bilinear integral and its one-dimensional moments are

$$\mathcal{B}(F,G)=\int_0^1\int_0^1\frac{F(s)G(t)}{\sqrt{1-s^2t^2/4}}\,ds\,dt,\quad J_j=\int_0^1\frac{t^j\,dt}{\sqrt{1-t^2/4}},\quad I_j=\int_0^1\frac{-t^j\log t\,dt}{\sqrt{1-t^2/4}}.$$

Here $I_0=V$, and integration by parts gives

$$I_j=\frac{4(j-1)}{j}I_{j-2}-\frac{4}{j}J_{j-2}+\frac{1}{j}J_j\qquad(j\ge2\text{ even}).$$

For even i, the entries $M_{ij}=\mathcal{B}(t^i,t^j)$ have the moment algebra

$$J_j=\frac{\pi}{3}c_j+\sqrt{3}\,d_j(1)-2d_j(0),\qquad M_{ij}=\begin{cases}(J_j-J_i)/(i-j), & i\ne j,\\ I_i, & i=j,\end{cases}\qquad[V]I_i=c_i.$$

It is generated by the recurrences in [5]; it is a formal coordinate convention, not an independence claim. For an increasing set $A = \{a_0, \dots, a_{n-1}\}$ of even indices in $\{0, \dots, 4n-1\}$, write the coordinate matrices on these rows and all $4n$ columns as R, T, P, C. Thus the specified moment entries are $R + \sqrt{3}T + \pi P + VC$.

Theorem 3.2. *For every such A, the square matrix $\mathcal{M}_A = (R^{\mathsf{T}}, T^{\mathsf{T}}, P^{\mathsf{T}}, C^{\mathsf{T}})^{\mathsf{T}}$ is invertible. More precisely,*

$$|\det \mathcal{M}_A| = 2^n 3^{n^2-n} C_n \frac{\mathcal{V}(A)^4 \prod_{a\in A} c_a}{\prod_{a\in A} a!(4n-1-a)!}. \tag{3.8}$$

For $A = (0, 2, \dots, 2n-2)$, its signed determinant is

$$\det \mathcal{M}_A = (-1)^n 2^{6n^2-3n} 3^{n^2-n} \left(\prod_{i=0}^{n-1} \frac{(i!)^4((2i)!)^2}{((2n+2i+1)!)^2} \right) \prod_{j=1}^{2n-1} j!. \tag{3.9}$$

In this case

$$\log|\det \mathcal{M}_A| = n^2 \log(3/16) + O(n \log(n+1)). \tag{3.10}$$

Proof. Let $D(z) = \prod_{a\in A}(z-a)$ and $O = \{0, \dots, 4n-1\} \setminus A$. If $\mathcal{M}_A q = 0$, then $Cq = 0$ forces $q_a = 0$ on A. Construct

$$F(t) = \sum_{j\in O} \frac{q_j}{D(j)} t^j + \sum_{a\in A} u_a t^a$$

with $\theta^k F(1) = 0$ for $k < n$, where $\theta = t\,d/dt$. The u_a exist uniquely by the Vandermonde determinant. For $L_a(z) = D(z)/(z-a)$,

$$L_a(\theta)F(t) = \sum_{j\in O} \frac{q_j}{j-a}(t^j - t^a).$$

Applying each coefficient functional $t^j \mapsto c_j, d_j(0), d_j(1)$ gives zero by $Pq = Rq = Tq = 0$. The L_a form a basis of polynomials of degree below n. Thus F is annihilated by all four blocks of $\mathcal{H}_n(1)$, which is nonsingular, so $q = 0$.

For completeness, keep determinants in this same change of variables. Let $\mathcal{A}$ be the $3n$ by $3n$ divided-difference matrix on columns O with row blocks

$$\frac{c_j - c_a}{a-j}, \qquad \frac{d_j(0) - d_a(0)}{a-j}, \qquad \frac{d_j(1) - d_a(1)}{a-j}.$$

Changing the four polynomial row bases from monomials to L_a has determinant $\mathcal{V}(A)^4$. Changing the coefficient variables of F to $(q_j)_{j\in O}$ and $(L_a(\theta)F)(1)$ has absolute determinant $\mathcal{V}(A)^2 |\prod_{j\in O} D(j)|$. Eliminating the last variables therefore gives

$$|\det \mathcal{A}| = |\det \mathcal{H}_n(1)|\, \mathcal{V}(A)^2 \Big/ \left| \prod_{j\in O} D(j) \right|.$$

The coordinate scalings are $-2, 1/3, 1$ on the $d(0), c, d(1)$ blocks, and the removed diagonal block contributes $\prod c_a$. Finally

$$\left| \prod_{j\in O} D(j) \right| = \frac{\prod_{a\in A} a!(4n-1-a)!}{\mathcal{V}(A)^2}.$$

This proves (3.8). For the stated even consecutive set, expansion along the last block contributes the sign $(-1)^{n(9n-3)/2}$. The coordinate scalings, the sign of $\det \mathcal{H}_n(1)$, and the product $\prod_{j\in O} D(j)$ contribute respectively $(-1)^n$, $(-1)^n$, and $(-1)^{n(n-1)/2}$. Their product is $(-1)^n$; substituting (3.3) gives (3.9). Stirling's estimate $\log(m!) = m\log m - m + O(\log(m+1))$, summed over the displayed products, cancels all $n^2 \log n$ terms. The remaining coefficient is $\log 3 - 4\log 2$, proving (3.10). □

3.5 The exact rational normalization

The small real size of the determinant and the size of its reduced numerator are very different. The following separates them precisely.

Theorem 3.3. *Define*

$$Q_n = \frac{\prod_{i=0}^{n-1}((2n+2i+1)!)^2}{\left(\prod_{i=0}^{n-1}(i!)^4((2i)!)^2\right)\left(\prod_{j=1}^{2n-1}j!\right)}.$$

Then Q_n is an integer. If the determinant in (3.9) is written as $(-1)^n a_n/b_n$ in lowest terms, with $a_n, b_n > 0$, then a_n has no prime factor other than 2 and 3, and

$$\log a_n = (6\log 2 + \log 3)n^2 + O(n\log(n+1)),$$
$$\log b_n = 10\log 2\, n^2 + O(n\log(n+1)).$$

For each fixed prime p, one also has $v_p(Q_n) = O_p(n\log(n+1))$.

Proof. For integers $q \geq 2$, put

$$F_n(q) = 4\sum_{i<n}\left\lfloor\frac{i}{q}\right\rfloor + 2\sum_{i<n}\left\lfloor\frac{2i}{q}\right\rfloor + \sum_{j<2n}\left\lfloor\frac{j}{q}\right\rfloor - 2\sum_{i<n}\left\lfloor\frac{2n+2i+1}{q}\right\rfloor.$$

We prove $F_n(q) \leq 0$. With $g = \gcd(2,q)$, summing the four arithmetic floor progressions gives the exact shift identity

$$F_{n+q}(q) - F_n(q) = -3q + 2g - 4 < 0. \tag{3.11}$$

For explicit verification, the increments of the four unweighted sums in their displayed order are

$$n, \qquad 2n + \frac{q+g-2}{2}, \qquad 4n+q, \qquad 6n + \frac{5q-g+2}{2}.$$

Thus it suffices to consider $0 \leq r < q$. Set

$$T_r = \sum_{k=1}^{3}\min\bigl(r, \max(0, 2r - \lfloor kq/2\rfloor)\bigr).$$

Counting the indices for which $2r + 2i + 1 \geq kq$ gives

$$F_r(q) = 2\max(0, r - \lceil q/2\rceil) + \max(0, 2r-q) - 2T_r.$$

For $r \leq q/2$ the two positive terms vanish. For $r > q/2$, the first term in T_r equals r. The sum of the two positive terms is $4r - 2q$ when q is even and $4r - 2q - 1$ when q is odd. Consequently $F_r(q) < 0$ in the latter range as well. This proves the claim for every n, q.

Legendre's factorial valuation formula now gives

$$v_p(Q_n) = -\sum_{k\geq 1} F_n(p^k) \geq 0$$

for every prime p, proving integrality. These same expressions give a quantitative bound: for $r < q$ one has $|F_r(q)| \leq 6r$, and (3.11) shows $|F_n(q)| \leq 11n$ for every $n \geq 1, q \geq 2$. Also $F_n(q) = 0$ for $q > 4n-1$. Therefore $v_p(Q_n) = O_p(n\log(n+1))$.

The determinant is exactly

$$(-1)^n \frac{2^{6n^2-3n}3^{n^2-n}}{Q_n}.$$

All numerator primes other than $2, 3$ are excluded. The fixed-prime valuation bounds show $v_2(a_n) = 6n^2 + O(n\log(n+1))$ and $v_3(a_n) = n^2 + O(n\log(n+1))$. This proves the numerator estimate. Subtracting (3.10) gives the denominator estimate. No assertion about an individual invariant factor or a target-specialized determinant follows from these product valuations. □

3.6 Consequences and exact scope

Four coordinate prescriptions for each of n rows require at least $4n$ coefficient variables. Theorem 3.2 attains that minimal column count. In particular, it gives unique rational polynomials of degree below $4n$ realizing any prescribed four-coordinate output. Rational target-free dual columns with actual integral outputs δ_{ij} exist in this window. Neither their signs nor small integer normalizations follow.

If $W(s) \neq 0$ is an even rational polynomial of degree $2m$, consider the rows $s^{2i}W(s)$, $0 \leq i < n$. They have full four-coordinate row rank $4n$ on columns of degrees $0, \ldots, 4(n+m)-1$. Indeed their coefficient matrix inside the $n+m$ ordinary even rows has rank n; apply its four diagonal copies to the invertible matrix of Theorem 3.2. No positivity or root-location hypothesis on W is needed. This does *not* settle nuisance normality in the smaller original $3n$-column weighted window.

Formula (3.9) also shows that the complete coordinate map is invertible over $\mathbb{Z}_{(p)}$ for every prime $p > 4n$: its entries are integral and its determinant is a unit. It does not determine its Smith factors at smaller primes. A small real determinant of a rational coordinate matrix is not an irrationality criterion; denominators, target specialization and lattice shape are separate issues.

4 Constructing the complete moment inverse

4.1 The map and the theorem

Let $V = \mathrm{Cl}_2(\pi/3)$. Define formal moments

$$J_0 = \pi/3, \qquad J_1 = 4 - 2\sqrt{3}, \qquad I_0 = V,$$
$$J_j = 4(j-1)J_{j-2}/j - 2\sqrt{3}/j,$$
$$I_i = 4(i-1)I_{i-2}/i - 4J_{i-2}/i + J_i/i \quad (i \geq 2 \text{ even}).$$

For even i, set $M_{ij} = (J_j - J_i)/(i-j)$ off the diagonal and $M_{ii} = I_i$. The matrix $\mathcal{M}_n$ has columns $0 \leq j < 4n$, rows $i = 0, 2, \ldots, 2n-2$, and coordinate blocks $(\beta, \tau, \sigma, \alpha)$ in the formal basis $(1, \sqrt{3}, \pi, V)$. Formal coordinates do not assume arithmetic independence of these numbers. These moments represent the double integrals with integrand $s^i t^j / \sqrt{1 - s^2t^2/4}$ on the unit square.

Write Ω_m for the odd part of $\mathrm{lcm}(1, \ldots, m)$, with $\Omega_0 = 1$, and put $L_p = \lfloor \log_p(2n-1) \rfloor$ and $L_p^0 = \lfloor \log_p(2n-2) \rfloor$, the latter defined as 0 for $n = 1$.

Theorem 4.1. *The matrix $\mathcal{M}_n$ has the explicit inverse constructed below. For a column z of that inverse:*

1. *if its input is a unit β coordinate, its common odd denominator divides Ω_{2n-2};*
2. *for a unit τ coordinate and $p \geq 5$, every entry has valuation at least $-L_p^0$;*
3. *for a unit σ coordinate and $p \geq 5$, every entry has valuation at least $-2L_p$;*
4. *for a unit α coordinate and $p \geq 5$, every entry has valuation at least $-4L_p$.*

Consequently the common inverse denominator away from 2 and 3 divides $\mathrm{lcm}(1, \ldots, 2n-1)^4$.

The target remains unresolved. Round61 proved exponential-height rational lifts and the weaker all-column bound $-8L_p$ [13]. The new ingredient is a primitive-level cancellation, not a new lattice-minimum bound. Heisei round 48v2 concerns a different odd-only output map [20]; its sharp binary denominator and removed prime band are not transferred here.

4.2 A self-contained endpoint reduction

Put $\theta = t\partial_t$, $c_{2r} = \binom{2r}{r}$, $c_{2r+1} = 0$, and

$$d_0(x) = 0, \quad d_1(x) = -2, \qquad d_j(x) = 4(j-1)d_{j-2}(x)/j - 2x^{j-1}/j.$$

For $F = \sum f_j t^j$ set $\ell_c(F) = \sum f_j c_j$, $\ell_0(F) = \sum f_j d_j(0)$ and $\ell_1(F) = \sum f_j d_j(1)$. Direct induction gives

$$J_j = \pi c_j/3 + \sqrt{3}d_j(1) - 2d_j(0), \qquad (4-x^2)d_j'(x) - xd_j(x) = 2(x^j - c_j), \tag{4.1}$$

$$d_{2r+1}(0) = -2\,16^r(r!)^2/(2r+1)!. \tag{4.2}$$

Let $D_n(z) = \prod_{h=0}^{n-1}(z-2h)$ and $L_i(z) = D_n(z)/(z-2i)$. If $X \in \mathbb{Q}[t]_{<4n}$ vanishes to order n at 1, then $q = D_n(\theta)X/n!$ has $\alpha = 0$ and

$$\beta_i(q) = 2\ell_0(L_i(\theta)X)/n!, \quad \sigma_i(q) = -\ell_c(L_i(\theta)X)/(3n!), \quad \tau_i(q) = -\ell_1(L_i(\theta)X)/n!. \tag{4.3}$$

Indeed $q_{2i} = 0$, and $L_i(\theta)X(1) = 0$ eliminates the subtracted J_{2i} in the divided difference.

For $\deg R < 2n$, use the parity Rodrigues operator

$$(\mathcal{A}_n R)(t) = \frac{t(4-t^2)^{n-1/2}}{n!}\,\partial_x^n\big((4-x)^{1/2}x^{n-1/2}R(\sqrt{x})\big)\Big|_{x=t^2}. \tag{4.4}$$

It is polynomial and dyadic rational, since

$$\mathcal{A}_n(t^r) = \sum_{k=0}^{n}(-1)^k\binom{1/2}{k}\binom{n-1/2+r/2}{n-k}t^{r+2k}(4-t^2)^{n-k}. \tag{4.5}$$

Its image is exactly the joint kernel of $\ell_c(\theta^k F)$ and $\ell_0(\theta^k F)$, $0 \le k < n$. To verify this, multiply its coefficient at t^{r+2j} by the relevant parity weight c_{r+2j} or $d_{r+2j}(0)$. The result is a fixed constant times $(-1)^j\binom{n}{j}$, so every degree-below-n moment vanishes. The leading coefficients $(-1)^n\binom{n+r/2}{n}$ are nonzero; dimension counting, using two ordinary parity Vandermonde matrices, gives equality.

For prescribed $\beta = b$, $\sigma = s$, choose $F \in \mathbb{Q}[t]_{<2n}$ by

$$F_{2i} = -3n!s_i/(c_{2i}D_n'(2i)), \tag{4.6}$$

$$F_{2j+1} = \frac{n!}{2d_{2j+1}(0)}\sum_{i=0}^{n-1}\frac{b_i\Lambda_j(2i)}{D_n'(2i)}, \qquad \Lambda_j(z) = \prod_{h\neq j}\frac{z-(2h+1)}{2(j-h)}.$$

Lagrange interpolation gives $\ell_c(L_iF) = -3n!s_i$ and $\ell_0(L_iF) = n!b_i/2$. Here and below L_iF means $L_i(\theta)F$.

We need one endpoint identity. Write $N = n-1$, $a_i = (-2)^N i!(N-i)!$, and

$$v_i(x) = \sum_{h=i}^{N}\binom{2h}{h}\binom{h}{i}(4-x)^{N-h}.$$

The adjoint identity, attributed to Heisei round43 [19], is

$$L_i(-1-\theta)(4-t^2)^{-1/2} = a_i v_i(t^2)/(4-t^2)^{n-1/2}. \tag{4.7}$$

For completeness, expand L_i in $D_k(z)$ and apply $D_k(-1-2x\partial_x)(4-x)^{-1/2} = (-2)^k(2k)!/[k!(4-x)^{k+1/2}]$; comparison gives the displayed v_i. Let $B_i(F)$ denote twice the boundary-at 1 term in integration by parts of $\int_0^1 L_iF/\sqrt{4-t^2}\,dt$, divided by $\sqrt{3}$. It is rational, depends only on the first N jets of F at 1, and is0 when those jets vanish. There is no boundary term at 0: in the expansion

in $t^k\partial_t^k$, the required derivatives of $t^k/\sqrt{4-t^2}$ of orders below k vanish there. Integrating (4.1) and then using (4.7) yields

$$\frac{\ell_1(L_iF)-B_i(F)}{a_i}=\frac{2\ell_0(L_iF)-\pi\ell_c(L_iF)/3}{\sqrt{3}a_i}+\frac{1}{n!}\int_0^1 v_i(x)U^{(n)}(x)\,dx, \tag{4.8}$$

whenever

$$U^{(n)}(x)=\frac{n!F(\sqrt{x})}{\sqrt{3}\sqrt{x}(4-x)^{n-1/2}}. \tag{4.9}$$

Lemma 4.2. *Suppose U satisfies (4.9), has rational jets at 1, and has finite derivatives through order $n-1$ at 0 lying in $(\mathbb{Q}+\pi\mathbb{Q})/\sqrt{3}$. Set*

$$R=-\operatorname{Tay}_{<2n,1}\left(t^{1-2n}\sqrt{\frac{3}{4-t^2}}\,U(t^2)\right),\quad X=F+\mathcal{A}_nR,$$

where $\operatorname{Tay}_{<2n,1}$ is the Taylor polynomial at 1 of degree below $2n$. Then X vanishes to order n at 1, and $\ell_1(L_iX)=0$ for every $i<n$.

Proof. Put $U_R(x)=x^{n-1/2}\sqrt{(4-x)/3}\,R(\sqrt{x})$. The first $2n$ jets at 1 of $U+U_R$ vanish. Equation (4.4) therefore makes the first n jets of X vanish. Integrating the last term of (4.8) by parts n times is legitimate: v_i has degree $n-1$, and the prescribed origin traces are finite. Its endpoint-at 1 contribution is rational. All origin terms and the first term on the right are in $(\mathbb{Q}+\pi\mathbb{Q})/\sqrt{3}$. The left side is rational. Since $1,\sqrt{3},\pi$ are linearly independent over $\mathbb{Q}$, those nonrational terms cancel and the left side equals just the endpoint-at 1 contribution. Apply this to $U+U_R$ and X. The endpoint jets and $B_i(X)$ vanish, proving the assertion. The traces of U_R at 0 through order $n-1$ are 0. □

4.3 Odd Laurent preimages

For $F=t^{2j+1}$, $0\le j<n$, take

$$U(x)=\frac{1}{\sqrt{3}}\sum_{h=0}^{j}\frac{(-1)^{n+h}\binom{j}{h}4^{j-h}(4-x)^{h+1/2}}{\binom{h+1/2}{n}}. \tag{4.10}$$

Differentiating n times proves (4.9). Its jets at 1 are rational; its origin traces are rational multiples of $1/\sqrt{3}$. More concretely, the expression whose Taylor polynomial is used in Lemma 4.2 is the Laurent polynomial

$$t^{1-2n}\sum_{h=0}^{j}\frac{(-1)^{n+h}\binom{j}{h}4^{j-h}(4-t^2)^h}{\binom{h+1/2}{n}}. \tag{4.11}$$

This gives an especially short exact construction for all β inverse columns.

4.4 Even centered-angle preimages

For $m\le 2n-2$, let $H_{n,m}(u)$ be the primitive, vanishing at 0, of $u^{2m}(1+u^2)^{n-m-2}$ obtained by the following finite rules:

$$u^{2m}(1+u^2)^{n-m-2}=\sum_{j=0}^{m}(-1)^j\binom{m}{j}(1+u^2)^{n-2-j}.$$

For nonnegative exponents expand binomially and integrate each even monomial. For exponent $-k<0$, use

$$\int \frac{du}{(1+u^2)^k} = A_k \arctan u + A_k \sum_{r=1}^{k-1} \frac{4^r}{2r\binom{2r}{r}} \frac{u}{(1+u^2)^r}, \tag{4.12}$$
$$A_k = \binom{2k-2}{k-1}/4^{k-1}.$$

This follows by one integration-by-parts recurrence, beginning with $k=1$. Write $H^{\circ}_{n,m}$ for the same expression with every $\arctan u$ replaced by $\arctan u - \pi/6$. For $F = t^{2j}$, $j<n$, define

$$U(x) = \frac{2n}{\sqrt{3}} \sum_{h=0}^{n-1} (-1)^h \binom{n-1}{h} 4^{j+h-n+1} x^{n-1-h} H^{\circ}_{n,j+h}\left(\sqrt{\frac{x}{4-x}}\right). \tag{4.13}$$

Without the circle this is exactly

$$\frac{n}{\sqrt{3}} \int_0^x (x-z)^{n-1} z^{j-1/2} (4-z)^{1/2-n}\, dz.$$

Indeed substitute $z = 4u^2/(1+u^2)$ after expanding $(x-z)^{n-1}$. Thus it satisfies (4.9) and has zero origin traces through order $n-1$. Centering subtracts only a polynomial of degree below n, times $\pi/\sqrt{3}$; the differential equation is unchanged. Its origin traces therefore have the form required in Lemma 4.2. At $x=1$ the centered angle vanishes, all algebraic terms normalized by $1/\sqrt{3}$ are rational, and all higher Taylor coefficients are rational. This completes the inverse construction with $\tau = 0$ for arbitrary β, σ inputs.

4.5 The remaining output blocks

For pure $\tau = u$, put $F=0$ and prescribe a polynomial U of degree below n by

$$y_i = -n! u_i / a_i, \qquad z_h = \frac{1}{\binom{2h}{h}} \sum_{i=h}^{N} (-1)^{i-h} \binom{i}{h} y_i,$$
$$[y^k] U(1+y) = \sum_{h=k}^{N} \binom{h}{k} (-3)^{h-k} n \binom{N}{h} z_h.$$

Set $R = \mathrm{Tay}_{<2n,1}(t^{1-2n}\sqrt{3/(4-t^2)}\,U(t^2))$ and $q = D_n(\theta)\mathcal{A}_n R/n!$; here the sign is positive. Formula (4.8) with zero origin traces shows the desired τ outputs. Explicitly the endpoint functional of $(4-x)^{N-h}$ is $\sum_{k=h}^{N} \binom{k}{h} 3^{k-h} [y^k] U(1+y)/(n\binom{N}{h})$; the displayed two Pascal inversions solve it. Since $U^{(n)} = 0$ and the first $2n$ jets are retained, $\mathcal{A}_n R$ has the required order-n zero at 1. The other output blocks vanish.

Finally, for a unit α_i output start with $q_0 = t^{2i}/c_{2i}$. Its α outputs are the desired unit vector. Its β outputs are 0. Subtract its τ, σ outputs using the already constructed columns. This proves invertibility without any unproved period relation or numerical rank test.

4.6 Arithmetic of the primitive construction

We record the elementary counting facts used below. For an odd prime p and $0 \le h < n$,

$$v_p\binom{h+1/2}{n} \le L_p, \qquad v_p\binom{2h}{h} \le L_p^0, \qquad v_p\big(2r\binom{2r}{r}\big) \le \lfloor \log_p(2r) \rfloor. \tag{4.14}$$

For the first identity write $a = h+1$, $b = n-h-1$ and use $\binom{h+1/2}{n} = (-1)^b \binom{2a}{a}\binom{2b}{b}/(4^n\binom{n}{a})$. At each odd prime-power level the two doubling carries minus the addition carry are at most1;

the odd factors in the defining falling product have absolute value at most $2n-1$. For the last identity, a prime-power dividing r contributes no doubling carry, so the valuation contribution per level is at most1. The central-binomial bound is the ordinary doubling-carry count. These arguments also prove the common-denominator, not merely individual-entry, assertions.

Generalized half-integer binomial coefficients are dyadic. So are $D_n(j)/n! = 2^n\binom{j/2}{n}$ and the coefficients of (4.5). Moreover $n!/D_n'(2i) = (-1)^{N-i}n\binom{N}{i}/2^N$ and $1/d_{2j+1}(0) = -(2j+1)\binom{2j}{j}/(2\,16^j)$. Finally $\Lambda_j(2i) = \binom{i-1/2}{j}\binom{n-i-1/2}{N-j}$ is dyadic. Thus the odd part of F for unit β input is dyadic, while for unit σ input its coefficients have valuation at least $-L_p$.

For odd F, (4.11) and (4.14) give a loss at most L_p for every odd p, with no exceptional prime 3. Taylor expansion of a Laurent monomial at 1 has integral coefficients. This improves to L_p^0: only when $2n-1=p^e$ is there a difference. In that case the only possible top-level pole in (4.11) is $h=j=n-1$. But F_{2n-1} is divisible by $2n-1$ in $\mathbb{Z}[1/2]$, by (4.6). That term is p-integral, and every other term loses at most $e-1$ powers. This proves part 1 of Theorem 4.1.

For even monomials, the algebraic coefficients of $H_{n,m}$ have odd denominator dividing Ω_{2n-2}: polynomial primitives use odd denominators at most $2n-3$, and the rational terms of (4.12) are controlled by (4.14). The angle coefficient is dyadic. To see precisely what happens to its local jets, write $x=1+y$ and

$$w(y)=\sqrt{\frac{1+y}{1-y/3}}, \qquad \sqrt{3}\,\frac{d}{dy}\left(\arctan\sqrt{\frac{1+y}{3-y}}-\frac{\pi}{6}\right)=\frac{1}{2\sqrt{1+y}\sqrt{1-y/3}}.$$

All coefficients on the right are integral away from 2 and 3. Integrating through Taylor degree $2n-1$ costs at most L_p. Also $\sqrt{3}u^{2r+1}=3^{-r}w^{2r+1}$ and $\sqrt{3}u/(1+u^2)^r=w(3-y)^r/4^r$. Thus every retained jet of (4.13) has valuation at least $-L_p$ for $p\ge 5$. Multiplication by $t^{1-2n}\sqrt{3/(4-t^2)}$, substitution $x=t^2$, Taylor truncation, and (4.5) introduce no further such prime. The initial F costs at most L_p, proving part 3.

For the pure τ construction only $1/\binom{2h}{h}$ introduces primes $p\ge 5$; all other displayed normalized coefficients are dyadic or have powers of 3 as denominators. This proves part 2.

For clarity we give the final α accounting rather than assuming cancellation. The recurrence for $d_j(1)$, $j\le 2n-2$, gives $v_p(d_j(1))\ge -L_p$: the explicit coefficients are interlaced products $-2\,4^r(j-1)(j-3)\cdots(j-2r+1)/(j(j-2)\cdots(j-2r))$, with at most one excess denominator multiple per odd prime-power level. In an off-diagonal entry of $q_0=t^{2i}/c_{2i}$, its τ coordinate therefore has valuation at least $-3L_p$, and its σ coordinate at least $-2L_p$. On the diagonal the normalized recurrence is

$$I_{2i}/c_{2i}=V+\sum_{\substack{2\le s\le 2i\\ s\ meven}}\frac{-4J_{s-2}+J_s}{sc_s}.$$

Its τ coordinate is again at least $-3L_p$. Its σ coordinate is $-(H_{2i}-H_i)/3$, since $4c_{s-2}/c_s = s/(s-1)$, and loses at most L_p. The correction by τ inverse columns costs at most one further L_p, and that by σ columns at most two. Both totals are at most $4L_p$; q_0 itself loses at most L_p. This proves part 4 and the theorem.

5 Three short-lcm powers suffice for the target inverse columns

Let $\mathcal{M}_n$ be the complete formal mixed-moment matrix with even row exponents $0,2,\dots,2n-2$, column exponents $0,\dots,4n-1$, and coordinate blocks $(\beta,\tau,\sigma,\alpha)$ for $(1,\sqrt{3},\pi,V)$. The Laurent and centered-angle construction gives its inverse. For $p\ge 5$, put

$$L=\lfloor \log_p(2n-1)\rfloor, \qquad C_a=\binom{2a}{a}.$$

The already established inverse-column bounds are $-L$ for a τ input and $-2L$ for a σ input. The source Meiji R62 [14] proved $-4L$ for the target input and explicitly left the apparent improvement to $-3L$ unproved. We now establish that improvement.

Theorem 5.1 (Improved target-column denominator)**.** *Every entry of a unit α column of $\mathcal{M}_n^{-1}$ has p-adic valuation at least $-3\lfloor \log_p(2n-1) \rfloor$ for every prime $p \geq 5$. Consequently, the common inverse denominator away from 2 and 3 divides*

$$\operatorname{lcm}(1, \ldots, 2n-1)^3.$$

The sharper earlier bounds for the other three input blocks remain valid. No optimality of the exponent three is asserted.

Lemma 5.2 (Shared carries)**.** *For distinct $0 \leq a, r < n$, the following valuations are at least $-L$:*

$$\frac{C_r}{(a-r)C_a}, \qquad \frac{C_r - C_a}{(a-r)C_a}.$$

Also $v_p(sC_s) \leq L$ for $1 \leq s < n$.

Proof. For every $h \leq n-1$, Legendre's factorial formula gives

$$v_p(C_h) = \sum_{j=1}^{L} \epsilon_j(h), \qquad \epsilon_j(h) = \lfloor 2h/p^j \rfloor - 2\lfloor h/p^j \rfloor \in \{0, 1\}.$$

If $k = v_p(a-r)$, then $k \leq L$ and $\epsilon_j(a) = \epsilon_j(r)$ for $j \leq k$. Write B for the sum of these common terms. Thus $v_p(C_a) \leq B + L - k$ and $v_p(C_r) \geq B$; also $v_p(C_r - C_a) \geq B$. Division by $(a-r)C_a$ proves both bounds. If $k = v_p(s)$, the first k doubling carries of s are zero. Hence $v_p(C_s) \leq L - k$, proving the last assertion. □

Write $d_{2a}(x)$ for the polynomial coefficient of the square-root period in the usual conic moment recurrence:

$$d_0(x) = 0, \qquad d_{2a}(x) = \frac{4(2a-1)}{2a} d_{2a-2}(x) - \frac{x^{2a-1}}{a}.$$

Induction, using $C_a/C_{a-1} = 2(2a-1)/a$, gives the exact identity

$$d_{2a}(x) = -C_a \sum_{s=1}^{a} \frac{x^{2s-1}}{sC_s}. \tag{5.1}$$

In particular $v_p(d_{2a}(1)) \geq v_p(C_a) - L \geq -L$.

Lemma 5.3 (Normalized nuisance bounds)**.** *Let $q_0 = t^{2a}/C_a$, where $0 \leq a < n$. For every even row $2r$, $0 \leq r < n$, its square-root and pi coordinates satisfy*

$$v_p(\tau_r(q_0)) \geq -2L, \qquad v_p(\sigma_r(q_0)) \geq -L.$$

Its rational coordinates vanish, and its target coordinates are the unit vector in position a.

Proof. For $r \neq a$, apart from p-adic units 2 and 3, the pi coordinate is $(C_a - C_r)/((r-a)C_a)$, so Lemma 5.2 applies. The square-root coordinate is

$$\frac{d_{2a}(1) - d_{2r}(1)}{2(r-a)C_a}.$$

In the polynomial version of this quotient, each coefficient with $s \leq \min(a, r)$ equals, up to sign and the unit 2,

$$\frac{C_r - C_a}{(a-r)C_a} \frac{1}{sC_s}.$$

Each coefficient with $a < s \le r$ equals $C_r/((a-r)C_a sC_s)$, up to a unit; each coefficient with $r < s \le a$ equals $1/((a-r)sC_s)$, up to a unit. The carry lemma bounds the first factor in every case below by $-L$, and $v_p(sC_s) \le L$ bounds the second below by $-L$. All polynomial coefficients therefore have valuation at least $-2L$, as does their sum at $x = 1$.

On the diagonal, the normalized recurrence reads

$$\frac{I_{2a}}{C_a} = V + \sum_{s=1}^{a} \frac{-4J_{2s-2} + J_{2s}}{2sC_s}.$$

Its pi coefficient is $-(H_{2a} - H_a)/3$, of valuation at least $-L$. Its square-root coefficient is

$$-\sum_{s=1}^{a} \frac{2d_{2s-2}(1) + 1}{2s^2 C_s}. \tag{5.2}$$

If $p \nmid s$, each summand has valuation at least $-2L$, by (5.1) and $v_p(C_s) \le L$. If $k = v_p(s) > 0$, put $c = v_p(C_s)$. The identity $C_s/C_{s-1} = 2(2s-1)/s$ gives $v_p(C_{s-1}) = c + k$, while the carry lemma gives $c + k \le L$. Equation (5.1) implies $v_p(d_{2s-2}(1)) \ge c + k - L$. This last number is nonpositive, so the numerator of the summand in (5.2) has valuation at least $c + k - L$. After division, its valuation is at least

$$(c + k - L) - (2k + c) = -L - k \ge -2L.$$

This proves the diagonal bound. The other formal coordinates follow directly from the mixed-moment recurrence: even J moments have no rational or target coordinate, and only the matching diagonal has target coefficient C_a. □

Proof of Theorem 5.1. Start the unit target inverse column with $q_0 = t^{2a}/C_a$. Its coefficient valuation is at least $-L$. Subtract its nuisance outputs using the established pure τ and pure σ inverse columns. Lemma 5.3 costs at most $2L$ and L for these outputs; the respective inverse columns cost at most L and $2L$. Both corrections therefore have all coefficient valuations at least $-3L$. Their finite sum has the same bound. There is no rational-output correction. This proves the target-column assertion and, together with the earlier block bounds, the common short-lcm denominator statement. □

Scope. This closes a specific open denominator question in the supplied corpus. It improves an exact rational inverse; clearing its denominator still multiplies the target linear forms. It does not produce the independent small integral controls required for an irrationality proof.

6 Sharp local period images and denominator thresholds

6.1 Canonical coordinates and statements

Let

$$\mathcal{B}(q, P) = \int_0^1 \int_0^1 \frac{q(s)P(t)}{\sqrt{1 - s^2t^2/4}}\, ds\, dt, \qquad V = \mathcal{B}(1, 1) = \mathrm{Cl}_2(\pi/3).$$

For even q, write canonically

$$\mathcal{B}(q, P) = \beta + \tau\sqrt{3} + \sigma\pi + \alpha V.$$

This means the rational coordinates defined by the recurrences below, not an assumption of linear independence of these constants. Write $\mathbb{Z}_{(p)}$ for the rationals integral at p.

Theorem 6.1 (Universal degree thresholds)**.** *Let $p \geq 5$ be prime, $q \in \mathbb{Z}_{(p)}[s]$ even with $\deg q < p$, and $P \in \mathbb{Z}_{(p)}[t]$.*

1. *If $\deg P < 2p$ and $\tau = 0$, then $\beta \in \mathbb{Z}_{(p)}$.*
2. *If $\deg P < 3p$ and $\tau = \sigma = 0$, then $p\beta \in \mathbb{Z}_{(p)}$.*

In both cases $\alpha \in \mathbb{Z}_{(p)}$. Neither assertion requires orthogonality, positivity, a Rodrigues representation, or any restriction on the coefficients beyond local integrality.

The first assertion was proved in round 28; it is restated and proved here. The second is the new degree specialization of round 13's moment law.

Theorem 6.2 (Exact unrestricted local image)**.** *Put $q_m(s) = (1 - s^2)^m$, where $m \geq 0$ and $2m < p$. For $3 \leq M < p^2$, define*

$$\Gamma_{m,M,p} = \{(\beta(P), \alpha(P)) : P \in \mathbb{Z}_{(p)}[t]_{\leq M},\ \tau(P) = \sigma(P) = 0\}.$$

Then

$$\Gamma_{m,M,p} = \begin{cases} \mathbb{Z}_{(p)}^2, & M < 2p, \\ p^{-1}\mathbb{Z}_{(p)} \times \mathbb{Z}_{(p)}, & 2p \leq M < 3p, \\ p^{-2}\mathbb{Z}_{(p)} \times \mathbb{Z}_{(p)}, & 3p \leq M < p^2. \end{cases}$$

In particular both thresholds of Theorem 6.1 are sharp, even for $m = 0$.

This is an unrestricted polynomial image theorem. It does not assert that an orthogonal or positive sublattice attains these images.

6.2 Moment formulas and the two pole layers

Set $f(t) = \sqrt{1 - t^2/4}$ and

$$J_j = \int_0^1 t^j / f(t)\, dt, \qquad I_j = \int_0^1 -t^j \log t / f(t)\, dt.$$

In the coordinate order $(1, \sqrt{3}, \pi, V, \Lambda)$, where $\Lambda = \log((2 + \sqrt{3})/4)$, the initial values and recurrences are

$$J_0 = (0, 0, 1/3, 0, 0), \quad J_1 = (4, -2, 0, 0, 0), \quad I_0 = (0, 0, 0, 1, 0), \quad I_1 = (4, -2, 0, 0, 4),$$

$$J_j = \frac{4(j-1)}{j} J_{j-2} - \frac{2}{j} e_{\sqrt{3}}, \qquad I_j = \frac{4(j-1)}{j} I_{j-2} - \frac{4}{j} J_{j-2} + \frac{1}{j} J_j.$$

The monomial moment is

$$M_{ij} = \mathcal{B}(s^i, t^j) = \begin{cases} (J_j - J_i)/(i - j), & i \neq j, \\ I_i, & i = j. \end{cases}$$

All these identities follow by integration by parts. Even first indices have zero logarithmic coordinate. The V coordinate is zero off the diagonal and equals $c_i = \binom{i}{i/2}$ on an even diagonal, proving the integrality assertion for α.

We recall, with proof, the needed special case of round 13. Put $\epsilon = (-1/p)$ and $\eta = (-3/p)$, where the parentheses denote Legendre symbols.

Lemma 6.3 (Low-first-index descent)**.** *For even $0 \le i < p$ and $j = ap + r < p^2$, $0 \le r < p$,*

$$(p^2 M^{\beta}_{ij}, p^2 M^{\tau}_{ij}, p M^{\sigma}_{ij}) \equiv \begin{cases} c_i(\epsilon M^{\beta}_{0a}, \eta M^{\tau}_{0a}, M^{\sigma}_{0a}), & r = i, \\ (0,0,0), & r \neq i \end{cases} \pmod p.$$

All scaled coordinates displayed here are p-integral.

Proof. Use formal series

$$A = (1-4z^2)^{-1/2} = \sum c_j z^j, \quad U = \int_0^z A(u)du, \quad T = \int_0^z \frac{A(u)}{1-u}du,$$

$$H = \tfrac{1}{3}\int_0^z \frac{1-A(u)}{u}du, \quad U_2 = \int_0^z U(u)du/u, \quad T_2 = \int_0^z T(u)du/u.$$

The recurrences give the generating functions

$$\sum J^{\beta}_j z^j = 4AU, \quad \sum J^{\tau}_j z^j = -2AT, \quad \sum J^{\sigma}_j z^j = A/3,$$

$$\sum I^{\beta}_j z^j = 4AU_2, \quad \sum I^{\tau}_j z^j = -2AT_2, \quad \sum I^{\sigma}_j z^j = AH.$$

In characteristic p, $A(z) = A_p(z)A(z^p)$ with $A_p = (1-4z^2)^{(p-1)/2}$, last coefficient ϵ and $A_p(1) = \eta$. The coefficients of U, T, U_2, T_2, H at $j > 0$ are respectively

$$c_{j-1}/j, \quad \Big(\sum_{k<j} c_k\Big)/j, \quad c_{j-1}/j^2, \quad \Big(\sum_{k<j} c_k\Big)/j^2, \quad -c_j/(3j).$$

Through degree $< p^2$, only $j = ap$ survives after scaling by p or p^2 respectively. The factorization of A therefore gives

$$pU \equiv \epsilon U(z^p), \quad pT \equiv \eta T(z^p), \quad p^2U_2 \equiv \epsilon U_2(z^p), \quad p^2T_2 \equiv \eta T_2(z^p), \quad pH \equiv H(z^p).$$

Multiplication by A proves the corresponding J, I residues and valuation bounds. If $r \neq i$, the moment divisor $i - j$ is a unit and the extra p kills the residue. If $r = i$ and $a > 0$, divide the J difference by $-ap$. The case $a = 0, r = i$ is a low integral diagonal and both sides are zero. This proves the lemma. □

The three high columns, with rows (β, τ, σ), are

$$W_1 = M_{01} = (-4, 2, 1/3)^T, \quad W_2 = M_{02} = (0, 1/2, -1/6)^T, \quad W_3 = M_{03} = (-32/9, 2, 1/9)^T. \tag{6.1}$$

Their first two nuisance columns have determinant $-1/2$, while

$$\det[W_1, W_2, W_3] = 2/9, \qquad \beta = -\tfrac{4}{3}\tau - 4\sigma \quad \text{on } \operatorname{span}(W_1, W_2). \tag{6.2}$$

For $\deg P < 3p$, set $\phi_a = \sum_{i \text{ even} < p} q_i P_{ap+i} c_i$, $a = 0, 1, 2$. Lemma 6.3 says that the scaled coordinates descend to $\sum \phi_a M_{0a}$. The column M_{00} has only a V coordinate. Thus $\tau = \sigma = 0$ forces $\phi_1 = \phi_2 = 0$, and then $p^2\beta \equiv 0$. This proves the second assertion of Theorem 6.1. More explicitly, throughout this degree range,

$$p\beta + \frac{4\epsilon}{3\eta}\, p\tau + 4\epsilon\sigma \in \mathbb{Z}_{(p)}. \tag{6.3}$$

6.3 The first threshold and low-degree surjectivity

For completeness we give the stronger first-threshold argument from round 28. Put $c_p = J_p^\beta / J_p^\tau$. The odd antiderivative at $p = 2h+1$ gives

$$J_p = 2\sum_{r=0}^{h} (-1)^r \binom{h}{r} 4^{h-r} \frac{2^{2r+1} - 3^r\sqrt{3}}{2r+1}.$$

Only $r = h$ has a p denominator, so $pJ_p^\beta \equiv 4\epsilon$ and $pJ_p^\tau \equiv -2\eta \neq 0$. In particular c_p is a unit. The recurrence for $A_j = J_j^\beta - c_p J_j^\tau$ is integral for $j < 2p$, with its only singular step removed by $A_p = A_0 = 0$. Comparing recurrence steps $p+i$ and i gives

$$A_{p+i} \equiv A_i \pmod{p} \qquad (i < p \text{ even}).$$

The only off-diagonal moment divisor divisible by p is $j = p+i$, so $\beta - c_p\tau$ is integral on every even $i < p, j < 2p$. Low diagonals are integral. This proves the first assertion.

We also need a low-degree surjectivity fact, again restating round 28's explicit boundary model. Let $H_m(t) = \int_0^1 q_m(s)/f(st)\,ds$, and put $h_m = H_m(1)$, $h_m' = H_m'(1)$. Its differential operator and adjoint are

$$L_m = t(4-t^2)D^2 + (8m+8-3t^2)D - t, \quad L_m^* = t(4-t^2)D^2 - (8m+3t^2)D - t.$$

Integration by parts gives canonically

$$\mathcal{B}(q_m, L_m^* R) = [3R'(1) - (8m+4)R(1)]h_m - 3R(1)h_m' + \kappa_m R(0), \quad \kappa_m = 4^{m+1}/\binom{2m}{m}. \tag{6.4}$$

The V coefficient is checked directly by the binomial recurrence; the other three coordinates follow either from the moment recurrences or their known rational linear independence. The latter uses transcendence of π, not any unproved fact about V. The initial values $h_0 = \pi/3$, $h_1 = \sqrt{3} - \pi/3$, $h_0' = 2\sqrt{3}/3 - \pi/3$ and

$$h_m' = 2mh_{m-1} - (2m+1)h_m, \quad (m+1)h_{m+1} + (2m+1)h_m - 3mh_{m-1} = 0 \quad (m > 0)$$

show that

$$\det \begin{pmatrix} h_m^\tau & h_m^\sigma \\ (h_m')^\tau & (h_m')^\sigma \end{pmatrix} = -\tfrac{2}{9}(-3)^m.$$

This and κ_m are units when $2m < p$. Taking $R = (1-t)^2$ in (6.4) represents a rational unit. Quadratic Hermite interpolation with $R(0) = 0$ independently supplies arbitrary integral nuisance coordinates. Finally $\mathcal{B}(q_m, 1)$ has $\alpha = 1, \beta = 0$; its nuisance coordinates are integral and can be cancelled in this way. Consequently the degree-three output map, with all four coordinates, is an isomorphism over $\mathbb{Z}_{(p)}$. Its determinant in row order $(\beta, \alpha, \tau, \sigma)$ is explicitly

$$-\frac{2\,4^m(-3)^m}{9\binom{2m}{m}},$$

a unit. Thus degree-three polynomials realize every integral formal output.

6.4 Sharpness and proof of the local phase theorem

Write C_j for the full output of $\mathcal{B}(q_m, t^j)$, and retain only its three coordinates (β, τ, σ) when applying determinants. Since $q_m(0) = 1$ and $\deg q_m < p$, Lemma 6.3 gives

$$\operatorname{diag}(p^2, p^2, p)[C_p, C_{2p}, C_{3p}] \equiv \operatorname{diag}(\epsilon, \eta, 1)[W_1, W_2, W_3] \pmod{p}. \tag{6.5}$$

All three columns have $\alpha = 0$. For degree $2p$, the quotient $x = C^{\tau}_{2p}/C^{\tau}_{p}$ is integral and satisfies $x \equiv 1/4 \pmod{p}$. The polynomial

$$P_* = p(t^{2p} - xt^p)$$

has $\tau = 0$, integral σ, and $\alpha = 0$. Moreover

$$p\beta(P_*) \equiv \epsilon \neq 0 \pmod{p}.$$

Degree-three surjectivity supplies a correction with integral coefficients and output $(0, 0, 0, \sigma(P_*))$ in row order $(\beta, \alpha, \tau, \sigma)$. Subtract it. This gives a nuisance-free simple pole with zero α, proving attainment in the middle range.

For degree $3p$, solve the two scaled nuisance equations using C_p, C_{2p} to cancel those of C_{3p}. Their scaled determinant is $-\eta/2$, a unit. The resulting coefficients are in $\mathbb{Z}_{(p)}$, and the full determinant in (6.5) is $2\epsilon\eta/9$, also a unit. Hence the resulting nuisance-free output has $p^2\beta$ a unit and $\alpha = 0$. Together with the integral pair generators, these constructions give every claimed pair. The reverse inclusions follow from Theorem 6.1 in the first two ranges, and the pole-order bounds in Lemma 6.3 in the third. This proves Theorem 6.2. All witnesses can be made genuinely integral polynomials by multiplying by a common denominator prime to p; their exact pole orders are unchanged.

6.5 A common denominator and the improved Bernstein budget

Write $D_N = \operatorname{lcm}(1, \dots, N)$. For integral input polynomials of degrees at most N, the rational and square-root moment coordinates have denominator dividing $4D_N^2$. Here is a short verification. For odd and even indices respectively, the incomplete primitive $J_j(t) = \int_0^t u^j/f(u)du$ is

$$J_{2r+1}(t) = 2^{2r+2} \sum_{k=0}^{r} \frac{(-1)^k \binom{r}{k}}{2k+1} (1 - f(t)^{2k+1}),$$

$$J_{2r}(t) = \binom{2r}{r} J_0(t) + 2tf(t) \sum_{k=1}^{r} \frac{(-1)^k \binom{2r}{r-k}}{k} U_{k-1}(1 - t^2/2).$$

The polynomial $U_{k-1}(1 - z/2)$ is integral. Expanding these formulas gives coefficients with denominator dividing D_j. Now $I_j = \int_0^1 J_j(t)dt/t$, with $\int_0^1 (f(t) - 1)dt/t = -\Lambda - 1 + \sqrt{3}/2$ and $\int_0^1 t^a f(t)dt = J_a - J_{a+2}/4$; all resulting indices are at most j. This pays at most one further D_j and the factor 4. The off-diagonal moment divisor is already paid by the second D_N.

Corollary 6.4 (Explicit general denominator)**.** *Let q, P be integral, q even,* $\deg q \leq D$, $\deg P \leq M$, *and $\tau = \sigma = 0$. Put $N = \max(D, M, 1)$ and*

$$u = \max(3, D, \lfloor M/2 \rfloor), \qquad v = \max(3, D, \lfloor M/3 \rfloor),$$

$$G_{D,M} = \frac{4D_N^2}{\displaystyle\prod_{u<p\leq N} p^2 \prod_{v<p\leq \min(u,N)} p}.$$

Products run over primes. Then $G_{D,M}\beta \in \mathbb{Z}$ and $\alpha \in \mathbb{Z}$. If $D/n \to \delta$, $M/n \to \mu$ with positive finite $\max(\delta, \mu)$, *then*

$$\frac{\log G_{D,M}}{n} \longrightarrow \max(\delta, \mu/2) + \max(\delta, \mu/3).$$

Proof. Every removed prime exceeds D and 3. In the first product $M < 2p$, so all of its denominator is removed. In the second $M < 3p$, so at most a simple pole remains. Also $N < 3p < p^2$ for every removed prime, so its original exponent in D_N is one and the displayed quotient is an integer. All other prime powers retain the global bound. The prime number theorem gives the asserted limit. □

7 Optimizing the Bernstein degree and denominator budget

The positive-series and saddle formulas are from Taisho R36. Combining them with the stronger universal degree denominator of Taisho R38 admits an optimization which was not performed in those source rounds. It distinguishes the best analytic and arithmetic balance from the unsolved problem of realizing that balance by positive integral controls. In particular, the uniform positive cone proved at $m/n = 3/20$ is not assumed to persist at the new optimal parameter.

Let n, m, K be nonnegative integers with $n > 0$, put

$$q_m(s) = (1-s^2)^m, \qquad P_{n,K,d}(t) = \frac{(-1)^n}{n!}\frac{d^n}{dt^n}\left[t^{n+d}(1-t)^{n+K-d}\right], \quad 0 \le d \le K,$$

and define $T_{n,m,K,d} = \mathcal{B}(q_m, P_{n,K,d})$. Every $P_{n,K,d}$ has integer coefficients. Write $T_{n,m,K} = T_{n,m,K,K}$ for the last column. For fixed real parameters $\rho > 0$ and $\kappa \ge 0$, define

$$\begin{aligned} F_{\rho,\kappa}(x) = {} & -x\log 2 + \frac{x}{2}\log\frac{x}{x+2\rho} + \rho\log\frac{2\rho}{x+2\rho} + x\log x - (x-1)\log(x-1) \\ & + (x+\kappa)\log(x+\kappa) - (x+\kappa+1)\log(x+\kappa+1), \qquad x > 1, \end{aligned} \tag{7.1}$$

$$\chi(\rho,\kappa) = -\max_{x>1} F_{\rho,\kappa}(x), \tag{7.2}$$

$$g(\rho,\kappa) = \max\left(2\rho, \frac{1+\kappa}{2}\right) + \max\left(2\rho, \frac{1+\kappa}{3}\right), \tag{7.3}$$

$$B(\rho,\kappa) = \chi(\rho,\kappa) - g(\rho,\kappa). \tag{7.4}$$

Theorem 7.1 (A sharp budget optimization within this family)**.** *The following statements hold.*

1. *If $m/n \to \rho > 0$ and $K/n \to \kappa \ge 0$, then*

$$0 < T_{n,m,K,d} \le T_{n,m,K} \quad (0 \le d \le K), \qquad \log T_{n,m,K} = -\chi(\rho,\kappa)n + o(n).$$

   When $m = \rho n + O(1)$ and $K = \kappa n + O(1)$, the last error is $O_{\rho,\kappa}(\log n)$.

2. *For each fixed $\kappa \ge 0$, the function $\rho \mapsto B(\rho,\kappa)$ has a unique maximum, at*

$$\rho = \frac{1+\kappa}{6}.$$

3. *The optimized function*

$$B_*(\kappa) = B\left(\frac{1+\kappa}{6}, \kappa\right)$$

   is concave and strictly decreasing on $[0,\infty)$. Its endpoint value is

$$B_*(0) = 1.351404651210758218\ldots.$$

   In particular, for full width $K = n$, the optimal binomial order is $m = n/3 + O(1)$ and

$$B_*(1) = 1.175542931127682114\ldots.$$

   These maxima concern the explicitly defined budget (7.4), not all possible denominator estimates or approximation constructions.

Proof. Integrating Rodrigues' formula by parts, and using the positive kernel series, gives the exact expansion

$$T_{n,m,K,d} = \sum_{r \ge \lceil n/2 \rceil} \frac{\binom{2r}{r}}{16^r} \frac{4^m (2r)!\, m!\, (r+m)!}{r!\,(2r+2m+1)!} \frac{(2r)!\,(2r+d)!\,(n+K-d)!}{n!\,(2r-n)!\,(2r+n+K+1)!}. \tag{7.5}$$

All terms are strictly positive. At fixed r, the d-dependent factor $(2r+d)!(n+K-d)!$ is log-convex. Its value at $d = K$ is at least its value at $d = 0$, because

$$\frac{(2r+K)!n!}{(2r)!(n+K)!} = \prod_{j=1}^{K} \frac{2r+j}{n+j} \geq 1.$$

The largest endpoint therefore bounds every column term by term.

With $x = 2r/n$, Stirling's bounds applied to the last-column summand in (7.5) give $nF_{\rho,\kappa}(x) + O(\log n)$ on each compact subinterval of $x > 1$, when the parameter errors are $O(1)$. The derivatives needed to control its maximum are

$$F_x = -\log 2 + \frac{1}{2}\log\frac{x}{x+2\rho} + \log\frac{x}{x-1} + \log\frac{x+\kappa}{x+\kappa+1},$$
$$F_{xx} = \frac{\rho}{x(x+2\rho)} - \frac{1}{x(x-1)} + \frac{1}{(x+\kappa)(x+\kappa+1)}.$$

For $1 < x < 2$, the second expression is bounded above by

$$\frac{1}{2x} - \frac{1}{x(x-1)} + \frac{1}{x(x+1)} = \frac{x^2-5}{2x(x^2-1)} < 0.$$

Also $F_x \to +\infty$ as $x \downarrow 1$. For $x \geq 2$,

$$F_x < -\log 2 + \log\frac{x}{x-1} \leq 0.$$

Thus there is exactly one maximizer $x_{\rho,\kappa} \in (1,2)$, it is nondegenerate, and it is determined by

$$x^3(x+\kappa)^2 = 4(x+2\rho)(x-1)^2(x+\kappa+1)^2. \tag{7.6}$$

For completeness, the tail of the positive sum needs no unproved uniform saddle assertion. The ratio of consecutive last-column summands is a rational function of r, n, m, K. As $n \to \infty$ with $x = 2r/n$ fixed, it converges to $\exp(2F_x(x))$. On $x \geq 2$ this is bounded by a constant strictly less than one for the fixed parameters; the bound also holds at infinity, where the ratio tends to $1/4$. Consequently the same geometric bound holds for the exact ratios for all sufficiently large n and $r \geq n$. On the remaining compact interval, the unique nondegenerate maximum and Stirling's inequalities give the usual two-sided positive Laplace bounds. This proves the asserted logarithmic asymptotic. Mere convergence of the degree ratios gives an $o(n)$ error by continuity, with the same bounds on compact parameter neighborhoods.

The denominator theorem for degrees $D = 2m$ and $M = n + K$ gives

$$\log G_{2m,n+K} = g(\rho,\kappa)n + o(n).$$

There is no claim here that this universal denominator is the exact primitive denominator on the orthogonal or positive sublattice. Smooth dependence of the unique saddle and the envelope rule give

$$\chi_\rho = \log\left(1 + \frac{x_{\rho,\kappa}}{2\rho}\right) > 0, \qquad \chi_\kappa = \log\left(1 + \frac{1}{x_{\rho,\kappa}+\kappa}\right). \tag{7.7}$$

Below $\rho = (1+\kappa)/6$, the function g is the constant $5(1+\kappa)/6$, so the budget strictly increases. At and above this corner, g has slope at least 2 with respect to ρ, while

$$\chi_\rho < \log\left(1 + \frac{6}{1+\kappa}\right) \leq \log 7 < 2.$$

The budget strictly decreases above the corner. This proves its unique maximization in ρ.

For fixed x, the Hessian of F in (ρ,κ) is diagonal with strictly positive entries

$$F_{\rho\rho}=\frac{x}{\rho(x+2\rho)},\qquad F_{\kappa\kappa}=\frac{1}{(x+\kappa)(x+\kappa+1)}.$$

A pointwise supremum of convex functions is convex. It follows that $\chi=-\sup_x F$ is concave jointly in (ρ,κ), and hence

$$B_*(\kappa)=\chi\left((1+\kappa)/6,\kappa\right)-5(1+\kappa)/6$$

is concave. By (7.7), its right derivative at zero is

$$B_*'(0+)=\frac{1}{6}\log(1+3x_0)+\log(1+1/x_0)-\frac{5}{6},\tag{7.8}$$

where x_0 is the unique root in $(1,2)$ of

$$3x^5=4(3x+1)(x-1)^2(x+1)^2.$$

The exact rational enclosure

$$1.345489509099647509 34<x_0<1.345489509099647509 35$$

and elementary rational logarithm bounds give

$$-0.008147<B_*'(0+)<-0.008145<0.$$

Concavity now implies strict decrease on the entire half-line. The two displayed budget constants follow from the same interval calculation, using (7.1). At $\kappa=1$ the saddle is enclosed by

$$1.410916909577287301 84<x_{1/3,1}<1.410916909577287301 85.$$

All the inequalities can be replayed with rational arithmetic. For example, reduce a positive rational to $y\in[1,2]$, set $z=(y-1)/(y+1)$, and bound the logarithm by

$$2\sum_{j=0}^{N-1}\frac{z^{2j+1}}{2j+1}\le\log y\le 2\sum_{j=0}^{N-1}\frac{z^{2j+1}}{2j+1}+\frac{2z^{2N+1}}{(2N+1)(1-z^2)}.$$

This certifies the signs and decimal enclosures independently of floating-point optimization. □

7.1 Including every shifted binomial first factor

The older shifted family of Taisho R24 allows $q_{a,m}(s)=s^{2a}(1-s^2)^m$. Let $a/n\to\eta\ge 0$ and $m/n\to\rho\ge 0$. With $0\log 0=0$, replace the first-factor part of (7.1) by

$$h(x/2+\eta,\rho),\qquad h(y,r)=y\log y+r\log r-(y+r)\log(y+r).$$

More precisely, put

$$\begin{aligned}
\widetilde{F}_{\eta,\rho,\kappa}(x)=&\ -x\log 2+h(x/2+\eta,\rho)+x\log x-(x-1)\log(x-1)\\
&+(x+\kappa)\log(x+\kappa)-(x+\kappa+1)\log(x+\kappa+1),\\
\widetilde{\chi}(\eta,\rho,\kappa)=&\ -\max_{x>1}\widetilde{F}_{\eta,\rho,\kappa}(x),\\
\widetilde{g}(\eta,\rho,\kappa)=&\ \max\left(2(\eta+\rho),\frac{1+\kappa}{2}\right)+\max\left(2(\eta+\rho),\frac{1+\kappa}{3}\right),\\
\widetilde{B}=&\ \widetilde{\chi}-\widetilde{g}.
\end{aligned}$$

The same positive-series argument applies: its first-factor moment is now

$$\int_0^1 s^{2r+2a}(1-s^2)^m\,ds = \frac{1}{2}\mathrm{B}(r+a+1/2, m+1).$$

Thus $\widetilde{B}$ is precisely the analogue of the preceding universal-denominator budget for the entire shifted family.

Theorem 7.2 (Global optimum with shifts included)**.** *For every $\eta, \rho, \kappa \geq 0$,*

$$\widetilde{B}(\eta, \rho, \kappa) \leq B_*(0) - \log(3x_0)\eta + B_*'(0+)\kappa.$$

Consequently the unique global maximum of this budget is $B_*(0) = 1.351404651210758218\ldots$, attained at $(\eta, \rho, \kappa) = (0, 1/6, 0)$. For fixed full width $\kappa = 1$,

$$\widetilde{B}(\eta, \rho, 1) \leq B_*(1) + \log\left(\frac{2}{3x_1}\right)\eta, \qquad x_1 = x_{1/3,1},$$

and its unique maximum is attained at $(\eta, \rho) = (0, 1/3)$.

Proof. The entropy function $h(y, r)$ is convex on $y > 0, r \geq 0$. For $r > 0$ this follows from

$$\nabla^2 h = \frac{1}{y+r}\begin{pmatrix} r/y & -1 \\ -1 & y/r \end{pmatrix},$$

a positive semidefinite matrix, and the assertion at $r = 0$ follows by continuity. The remaining dependence of $\widetilde{F}$ on κ is convex, as before. Thus $\widetilde{\chi}$ is jointly concave; $\widetilde{g}$, a sum of maxima of affine functions, is convex. Their difference is concave on the nonnegative parameter octant.

The unique nondegenerate saddle still lies in $(1, 2)$: the first-factor contribution to F_x is now $\frac{1}{2}\log((x+2\eta)/(x+2\eta+2\rho)) \leq 0$, and its contribution to F_{xx} is at most $1/(2x)$. The earlier derivative argument applies, including $\rho = 0$ by continuity. In particular the envelope derivatives at either interior-in-ρ point used below are legitimate.

At $(\eta, \rho, \kappa) = (0, 1/6, 0)$, set

$$A = \log(1 + 3x_0), \qquad C = \log(1 + 1/x_0).$$

The gradient of $\widetilde{\chi}$ at that point is

$$\big(\log(1 + 1/(3x_0)),\ A,\ C\big).$$

Since $0 < A < \log 7 < 2$, the vector

$$(A, A, 5/6 - A/6)$$

is a subgradient of $\widetilde{g}$ there. Indeed the first maximum in $\widetilde{g}$ selects $(1+\kappa)/2$, and the tied second maximum can be averaged with weight $A/2$ on $2(\eta+\rho)$. The supporting-plane inequality for the concave difference therefore gives

$$\widetilde{B}(\eta, \rho, \kappa) \leq B_*(0) + \big[\log(1 + 1/(3x_0)) - A\big]\eta + \big[C - 5/6 + A/6\big]\kappa.$$

The two bracketed coefficients are respectively $-\log(3x_0)$ and $B_*'(0+)$. Both are strictly negative. If either $\eta > 0$ or $\kappa > 0$, the value is strictly below $B_*(0)$. Along $\eta = \kappa = 0$, the unique optimum in ρ was proved in Theorem 7.1.

For $\kappa = 1$, repeat the same supporting-plane calculation at $(\eta, \rho) = (0, 1/3)$, using $A = \log(1 + 3x_1/2) \in (0, 2)$. The ρ coefficient is zero and the η coefficient is $\log(2/(3x_1)) < 0$. The earlier unique optimum on $\eta = 0$ completes the proof. □

The optimizer $\kappa = 0$ is a boundary budget, not an existence theorem for positive cancelling controls. In particular, the single-column case $K = 0$ does not thereby acquire the two missing nuisance cancellations. Conversely, larger windows can improve the arithmetic existence of controls even while worsening this particular budget.

Corollary 7.3 (An improved full-width sufficient criterion)**.** *Suppose that, for infinitely many positive integers n divisible by 3, there are rational numbers $w_0, \dots, w_n \geq 0$, not all zero, such that*

$$P_w = \sum_{d=0}^{n} w_d P_{n,n,d} \in \mathbb{Z}[t], \qquad \tau(\mathcal{B}(q_{n/3}, P_w)) = \sigma(\mathcal{B}(q_{n/3}, P_w)) = 0,$$

and

$$\max_d w_d \leq \exp(cn) \quad \textit{for a fixed } c < 1.175542931127682114\dots.$$

Then V is irrational.

Proof. The universal denominator has rate $5n/3 + o(n)$, and the target coordinate α is integral because both input polynomials are integral. The positive series and the endpoint estimate give

$$0 < G_{2n/3,2n}\, \mathcal{B}(q_{n/3}, P_w) \leq \exp\big((c - B_*(1))n + o(n)\big) \longrightarrow 0.$$

If $V = a/b$ were rational in lowest terms, multiplying this positive number by b would give a positive integer tending to zero. □

The corollary permits rational Bernstein weights only when their actual Rodrigues output is coefficientwise integral. It does not turn a bounded positive rational kernel into an integral one without paying its true arithmetic cost. The positive-cone theorem at $(\rho, \kappa) = (3/20, 1)$ and its three-dimensional quotient bound therefore cannot simply be transferred to $(1/3, 1)$. The new conclusion closes the optimization of this universal budget. It does not close the infinite control-height problem or prove irrationality of V.

8 Complete first-layer saturation and the missing two-block interval

Let $L_j(t) = P_j(2t - 1)$ be the shifted Legendre polynomial, and put

$$\mathcal{O}_{n,K} = \{G \in \mathbb{Q}[t] : \deg G \leq n + K,\ \int_0^1 t^i G(t)\, dt = 0\ (0 \leq i < n)\}.$$

We distinguish the nested lattice and its coefficientwise saturation,

$$\Lambda = \sum_{j=0}^{K} \mathbb{Z} L_{n+j}, \qquad \Lambda^{\mathrm{sat}} = \mathcal{O}_{n,K} \cap \mathbb{Z}[t].$$

For a rational polynomial P, its canonical coordinates are specified by

$$P = P(0) + \frac{t}{2}\mathcal{D}U, \qquad \mathcal{D} = (4 - t^2)\partial_t^2 - 3t\partial_t - 1,$$

$$C(P) = (\alpha, \beta, \tau, \sigma) = \left(P(0), 2U(0), -U(1), \frac{3U'(1) - U(1)}{6}\right).$$

The diagonal coefficients of $\mathcal{D}$ are $-(j+1)^2$, so this specification is unambiguous. No independence of the associated real periods is assumed. For a coefficient lattice $\mathcal{L}$, let $\Gamma(\mathcal{L})$ be the pair image (β, α) of $F\mathcal{L}$ subject to the exact equations $\tau = \sigma = 0$.

The new point of this section is an *upper* bound for the whole saturated image. Selected divided columns give lower bounds, but do not by themselves show that further integral columns cannot enlarge the image. The complete reflected basis makes that distinction explicit.

8.1 The complete coefficient-selected source space

For a prime p, write

$$\mathcal{C}_p P(x) = \sum_{j\ge 0} \overline{[t^{jp}]P(t)}\, x^j.$$

The following lemma records the complete saturation mechanism from Reiwa R49 [26], with its proof for the present use.

Lemma 8.1. *Let $p > 3$, $d = n+K$, and $2\lfloor d/p\rfloor + 2 < p$. For each reflected pair $hp+l, hp+p-1-l \in [n,d]$, $0 \le l < (p-1)/2$, replace the larger nested column by*

$$D_{h,l} = \frac{L_{hp+p-1-l} - L_{hp+l}}{p}.$$

These columns form a basis of $\Lambda^{\mathrm{sat}}_{(p)}$. Moreover,

$$D_{h,l} \equiv (2h+1) L_h(t^p) D_{0,l}(t) \pmod p. \tag{8.1.1}$$

Here $D_{0,l}(0) = 0$, $\deg D_{0,l} = p-1-l$, and its leading coefficient is a p-unit.

Proof. The binomial expression

$$[t^j]L_N = (-1)^{N+j}\binom{N}{j}\binom{N+j}{j}$$

gives $L_{hp+r} \equiv L_h(t^p)L_r$ and $L_{p-1-l} \equiv L_l$ modulo p. This proves integrality of the divided columns. For completeness, put $\mathcal{L}_l = t(1-t)\partial_t^2 + (1-2t)\partial_t + l(l+1)$. The Legendre equations give

$$\mathcal{L}_l D_{h,l} \equiv (2h+1)(2l+1) L_h(t^p) L_l, \qquad \mathcal{L}_l D_{0,l} \equiv (2l+1) L_l.$$

Thus the difference in the asserted congruence is a polynomial solution of $\mathcal{L}_l Y = 0$. Every such solution is $L_l(t)A(t^p)$: the Wronskian equation gives $(Y/L_l)' = B(t^p)/(t(1-t)L_l^2)$; if $B \neq 0$, its leading Laurent term at zero has exponent congruent to -1 modulo p, impossible for a derivative. The roots of L_l are simple, so no pole of a rational function of t^p can be cancelled by L_l.

The coefficients of degrees jp, $0 \le j \le h$, fix A. Grouping factorials in complete p-blocks gives

$$[t^{jp}]L_{hp+r} \equiv (-1)^{h+r+j}\binom{h}{j}\binom{h+j}{j}(1 + 2jpH_r) \pmod{p^2}, \qquad H_r = \sum_{i=1}^{r} 1/i.$$

The block identity used here is $\binom{ap}{bp} \equiv \binom{a}{b} \pmod{p^2}$ for $a < p$; it follows by grouping nonmultiples of p and using $\sum_{i=1}^{p-1} 1/i \equiv 0 \pmod p$. Since $H_{p-1-l} \equiv H_l$ and the signs agree, all selected coefficients of $D_{h,l}$ vanish. Those of $L_h(t^p)D_{0,l}$ also vanish because $D_{0,l}(0) = 0$ and its degree is below p. As $L_l(0)$ is a unit, $A = 0$.

An ordinary column has reduced leading degree $hp + \min(r, p-1-r)$. Two such degrees coincide precisely for a reflected pair. After division by p, the new column has reduced leading degree $hp + p - 1 - l$: its leading binomial coefficient has exactly one low-digit carry and no high carry. All basis columns therefore have distinct reduced leading degrees and are independent modulo p. Their lattice is p-saturated in its rational span: reducing a relation $pG = \sum c_j B_j$ forces every c_j to be a multiple of p. This proves that the displayed lattice is the entire coefficient saturation. □

Theorem 8.2 (Complete first-layer source classification)**.** *Assume the hypotheses of Lemma 8.1. Write*

$$a = pu + s, \quad b = pv + w, \quad 0 \le s, w < p, \quad m = s + w < p/2,$$

$$F = (t-2)^a(t+2)^b, \qquad F_1(x) = (x-2)^u(x+2)^v.$$

Let

$$\mathcal{H} = \{h : [hp, (h+1)p-1] \cap [n, d] \neq \varnothing\}$$

and, if $m \geq 1$, *let*

$$\mathcal{J} = \{h \in \mathcal{H} : hp + m - 1 \geq n,\ (h+1)p - m \leq d\}.$$

Set $\mathcal{J} = \varnothing$ *when* $m = 0$. *Then the exact* $\mathbb{F}_p$*-linear image is*

$$\mathcal{C}_p(F\Lambda^{\mathrm{sat}}_{(p)}) = \operatorname{span}_{\mathbb{F}_p}\left(\{F_1 L_h : h \in \mathcal{H}\} \cup \{F_1 x L_h : h \in \mathcal{J}\}\right). \tag{8.2.1}$$

The nested image is exactly the span of the first set alone.

Proof. Put $F_0 = (t-2)^s(t+2)^w$. Frobenius gives $F \equiv F_1(t^p)F_0(t)$. Every ordinary column in block h reduces to $L_h(t^p)L_l(t)$ with $l \leq (p-1)/2$. The low product $F_0 L_l$ has degree below p and unit constant coefficient. Its selected image is thus a nonzero scalar multiple of $F_1 L_h$. This proves the nested assertion.

For a divided basis column, Lemma 8.1 gives the selected image of $(2h+1)F_1(t^p)L_h(t^p)F_0 D_{0,l}$. The low product has zero constant coefficient and degree below $2p$. Its selected polynomial is therefore $c_l x$ for a scalar c_l. It is zero whenever $l \geq m$, because then its degree is below p.

A reflected pair with $l < m$ is available only if $hp + m - 1 \geq n$ and $(h+1)p - m \leq d$. Conversely, whenever these inequalities hold, the pair with $l = m-1$ is available. Its low product has degree exactly p, with unit leading coefficient

$$\frac{1}{p}\binom{2(p-m)}{p-m};$$

there is exactly one carry because $p/2 < p - m < p$. The factor $2h+1$ is a unit by the size hypothesis. Thus it supplies $F_1 x L_h$. The full basis lemma now proves both inclusions in the claimed image, rather than only a lower bound. □

Corollary 8.3 (Only boundary blocks enlarge the first-layer image)**.** *In Theorem 8.2, suppose* $\mathcal{H} = \{H, H+1, \ldots, J\}$ *with* $H < J$. *Let* $\epsilon_- = 1$ *if* $H \geq 1$ *and* $H \in \mathcal{J}$, *and zero otherwise; let* $\epsilon_+ = 1$ *if* $J \in \mathcal{J}$, *and zero otherwise. Then the saturated selected image is exactly*

$$F_1 \operatorname{span}_{\mathbb{F}_p}\{L_{H-\epsilon_-}, \ldots, L_{J+\epsilon_+}\}.$$

Its dimension exceeds the nested dimension by precisely $\epsilon_- + \epsilon_+ \leq 2$, *irrespective of the much larger coefficient-saturation index.*

Proof. Use the three-term relation

$$xL_h = \frac{h+1}{2(2h+1)}L_{h+1} + \frac{1}{2}L_h + \frac{h}{2(2h+1)}L_{h-1}.$$

All denominators and the needed nonzero off-diagonal coefficients are units under the size hypothesis. An interior xL_h adds nothing. A boundary source supplies exactly its one missing neighbor; at $h = 0$ there is no lower neighbor. Distinct Legendre degrees in this short high range are independent over $\mathbb{F}_p$, and multiplication by F_1 is injective. □

If $\deg(FG) < p^2$, the elementary leading-residue identity is

$$(\alpha, p^2\beta, p^2\tau, p\sigma)(P) \equiv \operatorname{diag}(1, (-1/p), (-3/p), 1)C(\mathcal{C}_p P) \pmod{p}. \tag{8.2.2}$$

All four rows on the left are integral. This identity follows by separating multiples of p in the monomial integral recurrence

$$J_0 = \pi/3, \quad J_1 = 4 - 2\sqrt{3}, \quad jJ_j = 4(j-1)J_{j-2} - 2\sqrt{3},$$

and using $(J_0 - J_j)/j$ for the nonconstant monomial conic integral. Equivalently it follows from the resonant polynomials for $\mathcal{D}$; it is the same leading-residue identity used in Reiwa R49. Consequently Theorem 8.2 gives an exact finite-dimensional test for whether the whole scaled four-coordinate image is $\mathbb{Z}^4_{(p)}$: the listed high coordinate vectors must have rank four modulo p. Surjectivity modulo p lifts by a unit minor, and failure of rank four prevents surjectivity.

8.2 Closure of the two-block saturation gap

Specialize to

$$n = 100k, \quad K = 30k, \quad a = 21k, \quad b = k, \qquad F = (t-2)^{21k}(t+2)^k. \tag{8.2.3}$$

Keio R51 [25] proves the following nested-image input: there is a fixed nonzero integer polynomial $A(k)$ such that, for every compact interval $I \subset (19/100, 39/200)$, all sufficiently large k and all primes $p/n \in I$ with $p \nmid A(k)$,

$$\Gamma(\Lambda)_{(p)} = p^{-1}\mathbb{Z}_{(p)} \times p\mathbb{Z}_{(p)}. \tag{8.2.4}$$

Its lower inclusion uses the first nine nested columns together with FL_{6p+1}; its endpoint functional is J_2, not J_1. The present theorem adds the previously missing saturated upper inclusion.

Theorem 8.4 (Exact saturated equality on the two-block interval)**.** *With the same polynomial exceptions and compact-interval convention as the nested-image input, one has*

$$\Gamma(\Lambda^{\mathrm{sat}})_{(p)} = \Gamma(\Lambda)_{(p)} = p^{-1}\mathbb{Z}_{(p)} \times p\mathbb{Z}_{(p)} \qquad (19/100 < p/n < 39/200).$$

In particular saturation neither enlarges the pair image nor increases its least beta denominator at these primes.

Proof. Here $a = p + s$, $b = w$, $m = 22k - p < p/2$, and $\mathcal{H} = \{5, 6\}$. A nonzero divided source at $h = 5$ would require

$$4p + 22k - 1 \geq 100k,$$

which is impossible for $p < (39/2)k$. At $h = 6$ it would require

$$8p - 22k \leq 130k,$$

which is impossible for $p > 19k$. Thus $\mathcal{J}$ is empty. The entire saturated selected image is exactly the span of $q_5 = (x-2)L_5$ and $q_6 = (x-2)L_6$.

Their two high nuisance coordinates are independent, since the canonical triangular recurrence gives the exact determinant

$$\det \begin{pmatrix} \tau(q_5) & \sigma(q_5) \\ \tau(q_6) & \sigma(q_6) \end{pmatrix} = -\frac{150534053}{315} \neq 0.$$

Absorb its fixed numerator and denominator primes into A. For an exactly nuisance-free saturated input, the leading-residue identity therefore forces $\mathcal{C}_p P = 0$. The alpha row gives $\alpha \in p\mathbb{Z}_{(p)}$ and the beta row gives $\beta \in p^{-1}\mathbb{Z}_{(p)}$. This is the desired upper inclusion for *every* saturated input. The nested-image input supplies the reverse inclusion. □

Corollary 8.5 (No exponential gain from the closed gap)**.** *Let $\Lambda^{\mathcal{I}}$ be the partial saturation at all primes with $19n/100 < p < 39n/200$. Let G_n and $G_n^{\mathcal{I}}$ be the least positive integers clearing the beta coordinates after division by 2^{22k} on the nested and partially saturated nuisance-free pair images. Then*

$$\log[\Gamma(\Lambda^{\mathcal{I}}) : \Gamma(\Lambda)] = o(n), \qquad \log(G_n^{\mathcal{I}}/G_n) = o(n).$$

For either successive minimum and any common, possibly n-dependent, pair norm, the two genuinely normalized lattices have minimum ratio $\exp(o(n))$.

Proof. The Legendre expansion coefficient of an integer polynomial of degree at most $d = n + K$ is $(2j+1)\int_0^1 GL_j$. Consequently

$$B_d \Lambda^{\mathrm{sat}} \subseteq \Lambda, \qquad B_d = \operatorname{lcm}(1, \dots, 2d+1).$$

For proportional primes in question, $p^2 > 2d+1$ eventually; the rank-two pair index is thus at most p^2 at each prime. On every fixed compact subinterval, Theorem 8.4 leaves only primes dividing $A(k)$, whose total logarithmic contribution is $O(\log k)$. The two excluded boundary strips of relative width ε contribute at most $4\varepsilon n + o(n)$ by the prime number theorem. Let ε tend to zero after the limit in n.

Write the pair index as J_n and the denominator ratio as q_n. Then $q_n \mid J_n$, and for the normalized images Z, Z',

$$q_n Z \subseteq Z', \qquad J_n Z' \subseteq q_n Z.$$

Thus $(q_n/J_n)\lambda_i(Z) \le \lambda_i(Z') \le q_n \lambda_i(Z)$, which proves the claim for every common norm. □

This is a closed local-image and global-comparison question, not an irrationality assertion. It removes an unclassified candidate source of arithmetic improvement. It does not supply a small nonzero surviving form, nor bound the second minimum itself.

9 Quantitative arithmetic of the conic normalizers

The all-index normality theorems remove rational zeros from the high-block construction, but rational nonvanishing alone does not control the primes excluded when the high index grows. We close this quantitative gap. The argument uses only the polynomially specified coordinates, so it makes no assumption concerning rational independence of the real periods.

Put $B_d = \operatorname{lcm}(1, \dots, d)$, with $B_0 = 1$. Let $\rho(t) = 2/\sqrt{4-t^2}$. Define the density coordinates by the unique polynomial reduction

$$P = c + \tfrac{1}{2}((4-t^2)d'_P - t d_P), \qquad \mathcal{J}(P) = (-2d_P(0), d_P(1), c/3).$$

Define the conic coordinates by

$$P = P(0) + \tfrac{t}{2}((4-t^2)U''_P - 3tU'_P - U_P), \qquad \mathcal{I}(P) = \left(2U_P(0), P(0), -U_P(1), \frac{3U'_P(1) - U_P(1)}{6}\right).$$

We order these coordinates as $(\beta, \alpha, \tau, \sigma)$. For a reduced rational number $z = a/b$, $b > 0$, put $H(z) = \max(|a|, b)$; in particular $H(0) = 1$.

Lemma 9.1 (Uniform coefficient arithmetic)**.** *If $P \in \mathbb{Z}[t]$ has degree at most d and coefficient height at most $H \ge 1$, then*

$$3B_d \mathcal{J}(P) \in \mathbb{Z}^3, \qquad 3B_d^2 \mathcal{I}(P) \in \mathbb{Z}^4.$$

Moreover

$$\|\mathcal{J}(P)\|_\infty,\ \|\mathcal{I}(P)\|_\infty \le 14 H 2^d.$$

Consequently the reduced heights of all these coordinates are at most $42H2^dB_d^2 = H\exp(O(d))$, with an absolute implied constant.

Proof. For $j \ge 1$ the nonzero coefficients in d_{tj} are

$$[t^{j-1-2r}]d_{tj} = -2\,4^r \frac{(j-1)(j-3)\cdots(j-2r+1)}{j(j-2)\cdots(j-2r)}, \qquad 0 \le r \le \lfloor (j-1)/2 \rfloor.$$

For an odd prime p, and for each $p^a \le j$, the number of multiples of p^a in the denominator progression exceeds their number in the numerator progression by at most one. Indeed these are progressions of the same invertible step, of lengths $r+1$ and r; each count lies between the floor and ceiling of its length divided by p^a. Thus the denominator valuation is at most $\lfloor \log_p j \rfloor$. If j is odd the displayed denominator is odd. If $j = 2m$, its 2-adic valuation is bounded below by

$$r - v_2\big(m(m-1)\cdots(m-r)\big) = r - v_2((r+1)!) - v_2\binom{m}{r+1} \ge -\lfloor \log_2 m \rfloor.$$

Here $v_2((r+1)!) \le r$, and the factorial-floor formula bounds the binomial valuation by the number of powers of two at most m. Hence $B_j d_{tj}$ is integral. The remaining constant is $c(t^{2r}) = \binom{2r}{r}$ and $c(t^{2r+1}) = 0$, as follows either from the same triangular reduction or from the full density integral. This proves the assertion for $\mathcal{J}$.

Writing $\mathcal{J}_j = \mathcal{J}(t^j)$, direct reduction gives

$$\mathcal{J}_0 = (0, 0, 1/3), \quad \mathcal{J}_1 = (4, -2, 0), \quad \mathcal{J}_j = \frac{4(j-1)}{j}\mathcal{J}_{j-2} - (0, 2/j, 0) \quad (j \ge 2).$$

The conic rows satisfy $\mathcal{I}(1) = (0, 1, 0, 0)$ and, for $j > 0$,

$$\mathcal{I}(t^j) = \frac{1}{j}(-\mathcal{J}_j^{\beta}, 0, -\mathcal{J}_j^{\tau}, 1/3 - \mathcal{J}_j^{\sigma}).$$

Since $j \mid B_d$, this proves the second denominator assertion. The density recurrence gives $\|\mathcal{J}_j\|_\infty \le 6\,2^j$ by induction, and the last formula gives $\|\mathcal{I}(t^j)\|_\infty \le 7\,2^j$. Summing coefficients of P proves the displayed bounds. Finally $\log B_d = O(d)$, by the elementary Chebyshev bound for the least common multiple. This bound also follows by summing the central-binomial bound for prime powers over dyadic intervals. □

9.1 The normality mechanism

Here is a proof of the normality statements on which the fixed-window inversion and high-block arithmetic rest. It also specifies exactly why formal coordinates can be used without an arithmetic assumption about V.

Lemma 9.2 (One extra positive-density condition)**.** *Suppose a nonzero real polynomial Q is orthogonal on $(0,1)$ to polynomials of degree below k, and $\int_0^1 Q(t)\rho(t)\,dt = 0$. Then Q has at least $k+1$ sign changes in $(0,1)$.*

Proof. For $k = 0$ use positivity of ρ. If the number of sign changes is less than k, their product polynomial contradicts ordinary orthogonality. If it is exactly k, interpolate ρ at these nodes by a polynomial p of degree below k. The interpolation error has the sign of the node product, because $\rho^{(k)}(t) > 0$ on $(0,1)$. The latter follows from the positive power series for $(1 - t^2/4)^{-1/2}$. Thus $Q(\rho - p)$ has a fixed nonzero sign apart from finitely many zeros, contrary to its zero integral. □

Theorem 9.3 (Three density columns and four conic columns)**.** *For every $h \ge 1$ and every $n \ge 0$,*

$$\det\big(\mathcal{J}(L_{h-1}), \mathcal{J}(L_h), \mathcal{J}(L_{h+1})\big) \ne 0,$$
$$\det\big(\mathcal{I}(L_n), \mathcal{I}(L_{n+1}), \mathcal{I}(L_{n+2}), \mathcal{I}(L_{n+3})\big) \ne 0.$$

In these formulas the indicated coordinate vectors are the rows.

Proof. The density reduction gives the exact antiderivative

$$\rho P = c\rho + (\sqrt{4 - t^2}\, d_P)'.$$

Consequently $\mathcal{J}(P) = 0$ forces separately zero integrals of $P\rho$ over $(-2,0),(0,1),(1,2)$. If a nonzero combination Q of L_{h-1}, L_h, L_{h+1} had zero density coordinates, its ordinary orthogonality below degree $h-1$ and Lemma 9.2 would give h inner sign changes. Each outer interval adds one. This contradicts $\deg Q \le h+1$.

For the four-column assertion, suppose a nonzero combination Q of $L_n, \dots, L_{n+3}$ has $\mathcal{I}(Q) = 0$. Its alpha coordinate implies $Q(0) = 0$. Define

$$R(t) = \int_0^t Q(s)/s\, ds, \qquad Z(t) = R(t) - R(1).$$

The monomial identity in Lemma 9.1, including its constant $1/3$ term, gives exactly $\mathcal{J}(Z) = -\mathcal{I}_3(Q) = 0$, where $\mathcal{I}_3 = (\beta, \tau, \sigma)$. Also $Q = tZ'$ and $Z(1) = 0$. For $0 \le j < n$, integration by parts transfers the ordinary orthogonality:

$$0 = \int_0^1 t^j Q(t)dt = -(j+1)\int_0^1 t^j Z(t)dt.$$

The density lemma and the two outer intervals give $n+3$ distinct roots of Z in those open intervals; $Z(1) = 0$ gives one more. This contradicts $\deg Z \le n+3$. Since $Q \neq 0$, also $Z \neq 0$, so the contradiction is valid. □

Let $L_h(t) = P_h(2t-1)$ be the shifted Legendre polynomial and set

$$D_h = \det \begin{pmatrix} \mathcal{J}(L_{h-1}) \\ \mathcal{J}(L_h) \\ \mathcal{J}(L_{h+1}) \end{pmatrix} \quad (h \ge 1), \qquad T_h = \tau(tL_h),$$

$$E_h = \det \begin{pmatrix} \beta(L_h) & \beta(tL_h) \\ \tau(L_h) & \tau(tL_h) \end{pmatrix}, \qquad N_h = \det \begin{pmatrix} \tau(L_h) & \tau(tL_h) \\ \sigma(L_h) & \sigma(tL_h) \end{pmatrix}.$$

The all-index results of Keio R49–R50 [23, 24] give

$$D_h \neq 0, \qquad (-1)^h T_h > 0, \qquad (-1)^h E_h > 0, \qquad (-1)^h N_h > 0.$$

These hypotheses are used as established algebraic normality theorems, not as conclusions drawn from numerical approximation to the periods.

Theorem 9.4 (Uniform high-index heights and exceptional primes)**.** *There is an absolute constant $C > 0$ such that, for every $h \ge 1$, the reduced numerator and denominator heights of D_h, T_h, E_h, N_h are at most $\exp(Ch)$. In particular each of their absolute values is at least $\exp(-Ch)$.*

Let a_h, b_h, C_h be the unique rational normalizers determined by

$$w_h = \sigma + a_h\tau + b_h\beta, \quad w_h(L_h) = w_h(tL_h) = 0, \qquad C_h = \beta(tL_h)/T_h.$$

Their heights are at most $\exp(Ch)$ after increasing C. Write $v_h = \beta - C_h\tau$ and

$$S_h = \det \begin{pmatrix} w_h(\mathcal{J}(L_h)) & w_h(\mathcal{J}(tL_h)) \\ v_h(\mathcal{J}(L_h)) & v_h(\mathcal{J}(tL_h)) \end{pmatrix},$$

where the covectors act in density coordinate order (β, τ, σ). Then $S_h = D_h/(4(2h+1)T_h)$ is nonzero and has the same exponential height bound.

There is consequently a positive integer $\mathfrak{B}_H$ with

$$\log \mathfrak{B}_H = O(H^2)$$

such that for every $1 \le h \le H$ and every prime $p \nmid \mathfrak{B}_H$, all these normalizers are p-integral and D_h, T_h, E_h, N_h, S_h are p-units.

Proof. The coefficient identity

$$[t^j]L_h = (-1)^{h-j}\binom{h}{j}\binom{h+j}{j}$$

gives coefficient height at most 8^h. Lemma 9.1, with degree at most $h+1$, therefore bounds every coordinate needed here by reduced height $\exp(O(h))$. Determinants of fixed size and a fixed number of rational field operations preserve such a bound. Cramer's rule applies because $E_h \neq 0$, and C_h is defined because $T_h \neq 0$. The identity for S_h is the exact contact determinant identity of R49, followed by the two normalizing row operations; it also shows its nonvanishing. A nonzero reduced rational of height at most $\exp(Ch)$ has absolute value at least $\exp(-Ch)$.

For each h, multiply the nonzero numerators and denominators of D_h, T_h, E_h, N_h, S_h, and the denominators of a_h, b_h, C_h, taking absolute values. The product is a positive integer of logarithm $O(h)$. Let $\mathfrak{B}_H$ be the product of these integers for $1 \le h \le H$. Its logarithm is $O(\sum_{h\le H} h) = O(H^2)$, and its prime-divisor property is exactly the asserted one. □

Corollary 9.5 (Four-column rational inversion)**.** *Let A_h have rows $\mathcal{I}(L_h), \dots, \mathcal{I}(L_{h+3})$. The all-index four-column normality theorem implies*

$$\max_{i,j} H((A_h^{-1})_{ij}) \le \exp(O(h+1)).$$

Thus four consecutive columns realize any prescribed rational formal coordinate vector, with a rational control of height at most the input height to an absolute fixed power times $\exp(O(h+1))$.

Proof. Every entry of A_h has height $\exp(O(h+1))$ by Lemma 9.1. Its nonzero determinant and its cofactors use only a fixed number of rational arithmetic operations. The adjugate formula proves the bound. Multiplication by the prescribed vector is again a fixed number of rational operations. □

Corollary 9.6 (A growing-block exclusion budget)**.** *If $H = H(n) = o(\sqrt{n})$, all high-index normalizer exceptions for $1 \le h \le H$ can be removed at total prime logarithmic weight $o(n)$. More explicitly,*

$$\sum_{p\mid \mathfrak{B}_H} \log p = O(H^2), \qquad \#\{p \ge Y : p \mid \mathfrak{B}_H\} = O(H^2/\log Y) \quad (Y \ge 2).$$

Thus these particular exceptional primes do not alter a denominator exponent measured on the scale n in that growing-block range.

Proof. The sum is bounded by $\log \mathfrak{B}_H$. Each term with $p \ge Y$ is at least $\log Y$. Apply Theorem 9.4. □

This conclusion controls high rational normalizers only. It supplies neither the low-image rank needed for every new prime interval nor a short integral surviving form. In particular it cannot replace the second-successive-minimum estimate required for irrationality.

10 An exact dyadic index and an obstruction to short integral output bases

The direct odd output basis of the later Heisei manuscripts has exponential coefficient denominators. A different invariant measures the simultaneous cost of making an entire output basis integral. We compute its dyadic part exactly. This is a new consequence of the direct boundary-jet construction, rather than a claim about small forms containing the Gieseking volume.

Put

$$\mathcal{B}(f,g)=\int_0^1\int_0^1 \frac{f(s)g(t)}{\sqrt{1-s^2t^2/4}}\,ds\,dt, \qquad N=n-1, \qquad D_n(z)=\prod_{h=0}^{n-1}(z-2h).$$

For $0\le i<n$, define

$$f_i(x)=\frac{1}{(-2)^N(N-i)!}\sum_{h=0}^{i}\frac{(-1)^{i-h}(x-4)^h}{(2h)!(N-h)!(i-h)!}, \tag{10.1}$$

$$w(x)=(4-x)^{1/2}(1-x)^{2n}, \qquad Z_i(x)=[w^{-1}f_i]_{<n}, \qquad P_i(x)=(1-x)^{2n}Z_i(x), \tag{10.2}$$

$$U_nP=(4-x)^{n-1/2}\frac{d^n}{dx^n}\big((4-x)^{1/2}P\big), \tag{10.3}$$

$$G_{n,i}(t)=D_n(t\partial_t)\big[t(U_nP_i)(t^2)\big]. \tag{10.4}$$

The notation $[\,]_{<n}$ denotes Taylor truncation at zero. The square-root branch has constant term 2. All polynomials $G_{n,i}$ have rational coefficients, are odd, and have degree at most $6n-1$. In the coordinate $y=4-x$, the apparently algebraic operator is diagonal:

$$U_n(y^r)=(-1)^n(r+\tfrac12)_{\underline{n}}y^r. \tag{10.5}$$

Let C_n be their $(3n)\times n$ coefficient matrix, with rows indexed by $t,t^3,\dots,t^{6n-1}$, and let

$$\mathscr{W}_n=\operatorname{span}_{\mathbb{Q}}\{G_{n,0},\dots,G_{n,n-1}\}.$$

Lemma 10.1 (The output identities)**.** *For $0\le h,i<n$,*

$$\mathcal{B}(s^{2h},G_{n,i})=\delta_{hi}.$$

These values are rational by formal cancellation of the π and $\sqrt{3}$ coordinates; no linear independence involving the volume is assumed.

Proof. The explicit adjoint polynomial for row i is

$$h_i(x)=(-2)^Ni!(N-i)!\sum_{h=i}^{N}\binom{2h}{h}\binom{h}{i}(4-x)^{N-h}.$$

For any polynomial Z of degree less than n, set $F=wZ$ and $P=(1-x)^{2n}Z$. Direct integration of the odd monomial moments, followed by the adjoint identity for $D_n(z)/(z-2i)$, gives

$$\mathcal{B}\big(s^{2i},D_n(t\partial_t)[t(U_nP)(t^2)]\big)=-\int_0^1 h_i(x)F^{(n)}(x)\,dx.$$

Here the endpoint at 1 vanishes to order $2n$. Integrating by parts n times gives

$$\sum_{k=0}^{N}(-1)^kh_i^{(k)}(0)F^{(N-k)}(0).$$

There is no integral remainder because $\deg h_i<n$. In terms of $f=[F]_{<n}$ this boundary expression is

$$(Tf)_i=(-2)^Ni!(N-i)!\sum_{h=i}^{N}\binom{2h}{h}\binom{h}{i}(N-h)!f^{(h)}(4). \tag{10.6}$$

Upper triangular binomial inversion shows directly that the polynomials in (10.1) satisfy $Tf_i=e_i$. Since $[wZ_i]_{<n}=f_i$, this proves the asserted identities. The same calculation is the direct boundary-jet proof in Heisei R48 [20]; in particular it evaluates the actual integral. The formal odd moment expansion belongs to $\mathbb{Q}+\mathbb{Q}\pi+\mathbb{Q}\sqrt{3}$. Since $1,\pi,\sqrt{3}$ are linearly independent over $\mathbb{Q}$, its two nuisance coordinates vanish. Only this classical independence is used; the volume does not occur in these odd moments. □

Theorem 10.2 (Exact dyadic integral-output index)**.** *Let $s_2(j)$ be the number of ones in the binary expansion of j, with $s_2(0) = 0$. The lattice*

$$\Gamma_{n,2} = \{v \in \mathbb{Z}_2^n : C_n v \in \mathbb{Z}_2^{3n}\}$$

has exact index

$$[\mathbb{Z}_2^n : \Gamma_{n,2}] = 2^{A_n}, \qquad A_n = 4n^2 - 2n - 3\sum_{j=0}^{n-1} s_2(j). \tag{10.7}$$

Equivalently, the total negative Smith valuation of the coefficient map is A_n. All its Smith valuations are nonpositive. This does not assert an individual formula for every Smith factor.

Proof. We first compute the sum of its Smith valuations. The operator $D_n(t\partial_t)$ multiplies each odd monomial by a product of odd integers, so it is unimodular over $\mathbb{Z}_2$. By (10.5), $2^n U_n$ is unimodular in the y monomial basis. The coordinate change $y = 4 - x$ and its inverse are integral, so the same is true in the x basis. Multiplication by $(1-x)^{2n}$ from degree less than n to degree less than $3n$ is a saturated integral embedding: its first n coefficient rows form a unit lower triangular matrix. Thus its matrix is equivalent over $\mathbb{Z}_2$ to a column inclusion.

Let F_n denote the square coefficient matrix of $f_0, \dots, f_{n-1}$, and B_n the matrix of truncated multiplication by w^{-1} on polynomials of degree less than n. The preceding unimodular operations reduce the nonzero Smith valuations of C_n to those of

$$M_n = 2^{-n} B_n F_n.$$

The matrix B_n is lower triangular with diagonal $1/2$, hence $v_2(\det B_n) = -n$. In the $(x-4)$ monomial basis, F_n is upper triangular with diagonal

$$\frac{1}{(-2)^N (N-i)!^2 (2i)!}, \qquad 0 \le i < n.$$

The change to the x basis has determinant one. Consequently

$$\begin{aligned} v_2(\det M_n) &= -n^2 - n - nN - 2\sum_{i=0}^{n-1} v_2(i!) - \sum_{i=0}^{n-1} v_2((2i)!) \\ &= -4n^2 + 2n + 3\sum_{i=0}^{n-1} s_2(i) = -A_n, \end{aligned}$$

by $v_2(i!) = i - s_2(i)$ and $s_2(2i) = s_2(i)$.

A determinant alone would not identify the integral-preimage index if positive Smith valuations were present. We exclude that possibility using the actual rational-coordinate map. For the odd monomial t^{2r+1} its rational coordinate on row s^{2h} is

$$R_{h,r} = \frac{4\,16^r (r!)^2}{(2r+1)!(2h-2r-1)}. \tag{10.8}$$

For clarity, this follows by inserting $d_{2r+1}(0) = -2\,16^r (r!)^2/(2r+1)!$ into the formal moment identity

$$\mathcal{B}(s^{2h}, t^j) = \frac{J_j - J_{2h}}{2h - j}, \qquad J_j = \frac{\pi}{3} c_j + \sqrt{3} d_j(1) - 2 d_j(0),$$

where j is odd and $d_{2h}(0) = 0$. Every entry in (10.8) is 2-integral, since

$$v_2(R_{h,r}) = 4r + 2 - s_2(r) \ge 0.$$

Lemma 10.1 and its formal cancellation give $R_n C_n = I_n$. Thus C_n has an integral left inverse. In its Smith normal form a positive exponent would require a nonintegral entry in any left inverse. All exponents are therefore nonpositive. The index of the integral preimage is the product of 2 to their negatives, which is $2^{-v_2(\det M_n)} = 2^{A_n}$. □

The theorem concerns the output lattice, not merely a common denominator of one chosen basis. In particular it remains true after an integral unimodular change of output coordinates. Since $\sum_{j<n} s_2(j) = O(n\log(n+1))$, its logarithm is $4(\log 2)n^2 + O(n\log(n+1))$.

Combining the dyadic index with the odd-prime residue ranks. The sharp residue-rank theorem of Heisei R49 [21] supplies an independent contribution. We state its precise input here. For an odd prime $p > n$, all coefficient poles are at most simple. If $p \le 2n-2$, write $p = 2q+1$ and $n = q+1+a$. Then

$$r_{n,p} := \operatorname{rank}_{\mathbb{F}_p}(pC_n \bmod p) = \min\{a, \max(0, q-2a-1)\}; \tag{10.9}$$

outside that range $r_{n,p} = 0$. Thus the local integral-preimage index is $p^{r_{n,p}}$. This follows from the Taylor-tail map $r \mapsto (1-x)^d r - [(1-x)^d r]_{<n}$ on $\deg r < a$, where $d = p-1-2a$. Its rank is the right side of (10.9); the remaining two operators are injective on its tail image by the three root-multiplicity argument at $0, 1, 4$. No determinant-to-rank inference is involved.

Define the global lattice

$$\Gamma_n = \{v \in \mathbb{Z}^n : C_n v \in \mathbb{Z}^{3n}\}.$$

It has finite index because C_n is rational. Localizing and using Theorem 10.2 and (10.9) gives

$$\log[\mathbb{Z}^n : \Gamma_n] \ge \left(4\log 2 + \frac{1}{12}\right) n^2 + o(n^2). \tag{10.10}$$

Indeed, uniformly for $p > n$,

$$\frac{r_{n,p}}{n} = \psi(p/n) + O(1/n), \qquad \psi(t) = \begin{cases} \frac{3}{2}t - 2, & 4/3 \le t \le 3/2, \\ 1 - \frac{1}{2}t, & 3/2 \le t \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

The prime number theorem and partial summation give $\sum_{p>n} r_{n,p}\log p = n^2 \int \psi(t)dt + o(n^2) = n^2/12 + o(n^2)$. No assertion about the exact contribution of the remaining odd primes is needed for (10.10).

Corollary 10.3 (No uniformly subexponential integral output basis)**.** *Suppose $Q_{n,1}, \dots, Q_{n,n} \in \mathscr{W}_n \cap \mathbb{Z}[t]$ are linearly independent and all their rational output vectors*

$$v_j = (\mathcal{B}(1, Q_{n,j}), \mathcal{B}(s^2, Q_{n,j}), \dots, \mathcal{B}(s^{2n-2}, Q_{n,j}))$$

belong to $\mathbb{Z}^n$. For the coefficient norm $\|Q\|_1 = \sum_k |[t^k]Q|$,

$$\left(\prod_{j=1}^{n} \|Q_{n,j}\|_1\right)^{1/n} \ge \exp\left(\left(4\log 2 + \frac{1}{12} + o(1)\right) n\right). \tag{10.11}$$

In particular at least one control satisfies the same lower bound. The constant is $4\log 2 + 1/12 = 2.8559220555\ldots$.

Proof. By Lemma 10.1, the v_j are the coordinates of the $Q_{n,j}$ in the direct output basis. They are independent elements of Γ_n. Therefore their determinant has absolute value at least $[\mathbb{Z}^n : \Gamma_n]$. The integral kernel is at most $2/\sqrt{3}$ on the unit square, so

$$|\mathcal{B}(s^{2h}, Q)| \le \frac{2\|Q\|_1}{\sqrt{3}(2h+1)}, \qquad \|v_j\|_2 \le \frac{\pi}{\sqrt{6}}\|Q_{n,j}\|_1,$$

using $\sum_{h\ge 0}(2h+1)^{-2}=\pi^2/8$. Hadamard's inequality now gives

$$[\mathbb{Z}^n:\Gamma_n]\le\left(\frac{\pi}{\sqrt{6}}\right)^n\prod_{j=1}^{n}\|Q_{n,j}\|_1.$$

Insert (10.10) and take nth roots. □

This obstruction has a precise scope. It concerns a full set of n independent controls in the specified odd space, with both polynomial coefficients and rational outputs integral. It does not bound every single control, does not include arbitrary larger windows, and does not exclude small target-bearing forms. It shows that the exponential cost of a complete integral output basis cannot be removed merely by choosing a better rational basis. In contrast, a single small surviving form is exactly the arithmetic task that remains relevant to irrationality.

11 A calibrated one-row moment lattice

11.1 Moment coordinates and the assertion

Put

$$\mathcal{B}(F,G)=\int_0^1\int_0^1\frac{F(s)G(t)}{\sqrt{1-s^2t^2/4}}\,ds\,dt,\qquad V=\mathcal{B}(1,1)=\mathrm{Cl}_2(\pi/3).$$

The last equality follows by integrating in s, then putting $t=2\sin(u/2)$ and integrating by parts. In particular $1<V<6/5$. Write $J_j=\int_0^1 t^j(1-t^2/4)^{-1/2}\,dt$ and $I_j=\int_0^1 -t^j\log t\,(1-t^2/4)^{-1/2}\,dt$. The formal coordinate order is $(1,\sqrt{3},\pi,V)$, with

$$J_0=\pi/3,\qquad J_1=4-2\sqrt{3},$$
$$J_j=\frac{4(j-1)}{j}J_{j-2}-\frac{2}{j}\sqrt{3},\tag{11.1}$$
$$I_0=V,\qquad I_j=\frac{4(j-1)}{j}I_{j-2}-\frac{4}{j}J_{j-2}+\frac{1}{j}J_j\quad(j\ge 2\text{ even}).\tag{11.2}$$

These identities follow by integration by parts; differentiating the exponent in the first recurrence proves the second. Moreover

$$\mathcal{B}(s^i,t^j)=\frac{J_j-J_i}{i-j}\ (i\ne j),\qquad \mathcal{B}(s^i,t^i)=I_i.$$

All statements about coordinates below concern this specified formal expansion, not an assumed linear independence of the four constants.

Fix $a\ge 1$, put $m=2a$ and $C=\binom{2a}{a}$. Let $\mathcal{P}_m=t^m\mathbb{Q}[t]_{\le 3}$. Let $L_k(t)=P_k(2t-1)$ be the shifted Legendre polynomial. For $q=\sum_k u_kL_k$, define the rational norm

$$\mathcal{N}_m(q)=\frac{11}{10(m+1)}\sum_k|u_k|(5/29)^k.\tag{11.3}$$

Theorem 11.1. *The four-coordinate map $q\mapsto\mathcal{B}(s^m,q)$ is an isomorphism $\mathcal{P}_m\longrightarrow\mathbb{Q}^4$. Let $q_\beta,q_\alpha\in\mathcal{P}_m$ have formal outputs $1,V$, respectively, and set $\Lambda_m=\mathbb{Z}q_\beta+\mathbb{Z}q_\alpha$. For every real b,d,*

$$|b+Vd|\ \le\ \mathcal{N}_m(bq_\beta+dq_\alpha)\ \le\ 32|b+Vd|+\frac{104|d|}{C(m+1)^2}.\tag{11.4}$$

If $\lambda_2(m)$ is the second successive minimum of Λ_m for this norm, then

$$\lambda_2(2a)\longrightarrow 0\quad\Longleftrightarrow\quad V\notin\mathbb{Q}.$$

If $V=P/Q$ in lowest terms, $Q>0$, then $\lambda_2(2a)\ge 1/Q$ at every size.

11.2 A rational majorant independent of the target value

Lemma 11.2. *For every integer $m \ge 0$ and $k \ge 0$,*

$$0 < \mathcal{B}(s^m, L_k) \le \frac{\log 3}{m+1}(3+2\sqrt{2})^{-k} < \frac{11}{10(m+1)}(5/29)^k.$$

Consequently $|\mathcal{B}(s^m, q)| \le \mathcal{N}_m(q)$. If $q = t^m \sum_{j=0}^{3} c_j t^j$ and $H = \max_j |c_j|$, then

$$\mathcal{N}_m(q) \le \frac{8H}{(m+1)^2}. \tag{11.5}$$

Proof. Rodrigues' formula, with its boundary terms zero, gives

$$\int_0^1 t^r L_k(t)\,dt = \begin{cases} (r!)^2/((r-k)!(r+k+1)!), & r \ge k, \\ 0, & r < k. \end{cases}$$

Expand the positive kernel in even powers of st. Since $\binom{2r}{r}/16^r \le 4^{-r}$ and $(m+2r+1)^{-1} \le (m+1)^{-1}$, termwise comparison of these nonnegative Mellin moments bounds $\mathcal{B}(s^m, L_k)$ by $(m+1)^{-1}\int_0^1 L_k(t)/(1-t^2/4)\,dt$. For $z > 1$, another integration by parts gives

$$E_k(z) := \int_0^1 \frac{L_k(t)}{z-t}\,dt = \int_0^1 \left(\frac{t(1-t)}{z-t}\right)^k \frac{dt}{z-t}.$$

The maximum of the parenthesis is $(\sqrt{z} - \sqrt{z-1})^2$. Partial fractions and $t \mapsto 1-t$ give

$$\int_0^1 \frac{L_k(t)}{1-t^2/4}\,dt = E_k(2) + (-1)^k E_k(3) \le \log 3\,(3+2\sqrt{2})^{-k}.$$

Here $29/5 < 3 + 2\sqrt{2}$, and $\log 3 < 11/10$ because the degree-five Taylor sum of $e^{11/10}$ exceeds 3 by $15101/12000000$.

Finally $u_k = (2k+1)\int qL_k$ and $|L_k| \le 1$ on $[0,1]$, so $|u_k| \le 4H(2k+1)/(m+1)$. Since

$$\sum_{k\ge 0}(2k+1)(5/29)^k = 493/288, \qquad (11/10)\cdot 4 \cdot 493/288 < 8,$$

equation (11.5) follows. □

11.3 An explicit positive calibration

The recurrences give

$$[\pi]J_m = C/3, \quad [1]J_{m+1} = A := \frac{4\cdot 16^a (a!)^2}{(2a+1)!}, \quad [V]I_m = C, \quad [1]I_m = 0, \quad [\pi]I_m = -Ch/3,$$

where $h = H_{2a} - H_a \in (0,1)$. Set

$$r = \frac{4(m+1)}{m+2}, \quad \eta = \frac{4(m+2)}{m+3}, \quad P(t) = \frac{\eta}{\eta-1}t + \frac{2}{r-1}t^2 - \frac{3}{\eta-1}t^3,$$

$$Z(t) = \frac{-t+3t^3}{\eta-1}, \qquad W(t) = 1 - \frac{2h}{r-1}t^2.$$

Let $U = \mathcal{B}(s^m, t^m P)$, $T = \mathcal{B}(s^m, t^m Z)$ and $E = \mathcal{B}(s^m, t^m W/C)$. Direct coordinate substitution shows that $t^m P$ has only a $\sqrt{3}$ coordinate; $t^m Z$ has rational coordinate $-A$, zero π, V coordinates; and $t^m W/C$ has V coordinate one and zero rational and π coordinates.

For clarity, this calculation can also be done without guessing the polynomials. If $q = t^m(c_0 + c_1t + c_2t^2 + c_3t^3)$ has target coefficient d, then $c_0 = d/C$. Write $u = c_1 + c_3/3$ and $v = c_1 + \eta c_3/3$. Its rational coordinate is $-Av$, and its π coordinate vanishes exactly when $c_2 = 2(u - dh/C)/(r-1)$. Thus, after these three coordinates are fixed, the remaining freedom is exactly addition of $t^m P$.

The quadratic $P(t)/t$ is concave on $[0,1]$. Its endpoint values and $3 \le r < 4$, $16/5 \le \eta < 4$ imply

$$\frac{2}{3}t \le P(t) \le \frac{5}{2}t, \qquad |Z(t)| \le 2t, \qquad |W(t)| \le 2. \tag{11.6}$$

In particular $U > 0$. Its formal $\sqrt{3}$ coordinate is therefore nonzero, proving the asserted four-coordinate isomorphism. Writing $v = -1/A$, the unique rational polynomials with outputs $1, V$ satisfy the exact real identities

$$q_\beta = t^m \left(\frac{1 - vT}{U} P + vZ \right), \tag{11.7}$$

$$q_\alpha = t^m \left(\frac{V - E}{U} P + \frac{W}{C} \right). \tag{11.8}$$

Although these formulas display real constants, their coefficients are rational by the coordinate calculation just given. No rationality of V has been assumed.

The kernel lies between 1 and $6/5$. Therefore

$$\frac{2}{3(m+1)(m+2)} \le U \le \frac{3}{(m+1)(m+2)}, \quad |T| \le \frac{3}{(m+1)(m+2)}, \quad |E| \le \frac{3}{C(m+1)^2}. \tag{11.9}$$

Also $A \ge 4C$: the sequence $(2a+1)C^2/16^a$ starts at one and its consecutive ratio is $(2a+1)(2a+3)/(2a+2)^2 < 1$. The maximum coefficient heights of P, Z, W are at most $3/2, 3/2, 1$, respectively. Equations (11.9) yield

$$|T/U| \le 9/2, \quad |E/U| \le 6/C, \qquad H(t^{-m} q_\beta) \le 4(m+1)^2.$$

Subtracting V times (11.7) from (11.8) gives

$$q_\alpha - V q_\beta = t^m \left\{ \frac{W}{C} - \frac{E}{U} P + Vv \left(\frac{T}{U} P - Z \right) \right\}.$$

Its coefficient height is at most

$$\frac{1}{C} + \frac{9}{C} + \frac{6}{5} \frac{1}{4C} \left(\frac{27}{4} + \frac{3}{2} \right) = \frac{499}{40C} < \frac{13}{C}.$$

Applying Lemma 11.2 to these two polynomials proves (11.4), including its lower bound from their actual outputs.

11.4 The successive-minimum dichotomy

If $V = P/Q$, two independent integer pairs (b, d) cannot both lie in the one-dimensional line $Qb + Pd = 0$. One of their actual forms has absolute value at least $1/Q$, and hence so does one of their norms.

Conversely, suppose V is irrational. For every $\varepsilon > 0$ there exist two independent integer pairs with $0 < |b + Vd| < \varepsilon$. This is the elementary pigeonhole approximation argument: choose a primitive pair of sufficiently small nonzero error, and then a pair with still smaller nonzero error, which cannot be an integer multiple of the primitive pair. Fix these two pairs. Since $C(m+1)^2 \to \infty$, their norms have limsup at most 32ε by (11.4). Letting $\varepsilon \downarrow 0$ proves the converse and the theorem.

12 The exact Diophantine exponent of the calibrated moment lattice

This section strengthens the qualitative calibration dichotomy of the source Meiji R59 [12]. It determines exactly which Diophantine exponent is measured by that lattice. The result does not assume that the Gieseking constant is irrational; the rational alternative is included separately.

Put $V = \mathrm{Cl}_2(\pi/3)$, $m = 2a \geq 2$, and $C_m = \binom{2a}{a}$. Let $L_k(t) = P_k(2t-1)$ and write $q = \sum u_k L_k$. On $t^m\mathbb{R}[t]_{\leq 3}$ use the rationally defined norm

$$\mathcal{N}_m(q) = \frac{11}{10(m+1)} \sum_k |u_k|(5/29)^k.$$

The four formal-coordinate map for $\mathcal{B}(s^m, q)$ is invertible on this space. Denote its rational inverse columns for outputs $1, V$ by $q_{\beta,m}, q_{\alpha,m}$. The established calibration is

$$|b + Vd| \leq \mathcal{N}_m(bq_{\beta,m} + dq_{\alpha,m}) \leq 32|b + Vd| + \frac{104|d|}{C_m(m+1)^2}. \tag{12.1}$$

Moreover the coefficient of t^m of this polynomial is d/C_m: only the diagonal mixed moment has a target coordinate, equal to C_m. These are the only features of the moment construction used below. Let $\lambda_1(m) \leq \lambda_2(m)$ be the successive minima of $\mathbb{Z}q_{\beta,m} + \mathbb{Z}q_{\alpha,m}$ for $\mathcal{N}_m$.

For an irrational real number ξ, its irrationality exponent is

$$\mu(\xi) = \sup\{\nu : |\xi - p/q| < q^{-\nu} \text{ for infinitely many integers } p, q,\ q \geq 1\}.$$

The value $+\infty$ is permitted, and $1/\infty = 0$.

Theorem 12.1 (Quantitative calibration)**.** *For every even $m \geq 2$ and every real b, d, put $S_m(b,d) = \max\{|b + Vd|, |d|/C_m\}$. Then*

$$\frac{S_m(b,d)}{32(m+7)^{17}} \leq \mathcal{N}_m(bq_{\beta,m} + dq_{\alpha,m}) \leq 136 S_m(b,d). \tag{12.2}$$

If V is irrational, then

$$\limsup_{a\to\infty} \frac{-\log \lambda_1(2a)}{\log C_{2a}} = 1 - \frac{1}{\mu(V)}, \tag{12.3}$$

$$\liminf_{a\to\infty} \frac{-\log \lambda_2(2a)}{\log C_{2a}} = \frac{1}{\mu(V)}. \tag{12.4}$$

If instead $V = P/Q$ in lowest terms, $Q > 0$, the two ratios have limits 1 and 0, respectively, and $\lambda_2(2a) \geq 1/Q$ at every size.

The polynomial norm comparison. Write $q = t^m \sum_{j=0}^3 c_j t^j$ and $v_k = \int_0^1 q(t) t^k \, dt$ for $0 \leq k \leq 3$. Then

$$v_k = \sum_{j=0}^{3} \frac{c_j}{m+k+j+1}.$$

The 4 by 4 Cauchy matrix A_m in this identity has determinant

$$\det A_m = \frac{144}{\prod_{i,j=0}^3 (m+i+j+1)}.$$

Every cofactor has absolute value at most 6, by its six-term expansion and the fact that all entries have absolute value at most one. Hence every inverse entry is at most $(m+7)^{16}/24$. For $k \leq 3$, the

expansion of t^k in $L_0, \dots, L_k$ has nonnegative coefficients with sum one. Orthogonality therefore gives $|v_k| \le \sum_{\ell=0}^{3} |u_\ell|$. Consequently

$$\max_j |c_j| \le \frac{(m+7)^{16}}{6} \sum_{\ell=0}^{3} |u_\ell| \le \frac{10}{66}(29/5)^3 (m+7)^{17} \mathcal{N}_m(q) < 32(m+7)^{17} \mathcal{N}_m(q).$$

Apply this to $c_0 = d/C_m$ and combine with the lower bound in (12.1). The upper bound follows from the same calibration by replacing $(m+1)^{-2}$ by one. □

Lemma 12.2 (Two-dimensional minima product)**.** *For any real ξ and $0 < \varepsilon \le 1$, let*

$$S_\varepsilon(b,d) = \max\{|b + \xi d|, \varepsilon |d|\}, \qquad (b,d) \in \mathbb{Z}^2,$$

and let $s_1 \le s_2$ be its two successive minima. Then

$$\varepsilon/2 \le s_1 s_2 \le 2\varepsilon.$$

Proof. The image lattice $\{(b + \xi d, \varepsilon d) : (b,d) \in \mathbb{Z}^2\}$ has covolume ε. Two independent vectors of sup norms s_1, s_2 have determinant at least ε and at most $2 s_1 s_2$, proving the lower bound.

For completeness, $s_1 \le \sqrt{\varepsilon}$. Put $Q = \lfloor \varepsilon^{-1/2} \rfloor$. Among the $Q+1$ points $0, \xi, \dots, Q\xi$ modulo one, a circular gap is at most $1/(Q+1)$. This supplies $1 \le d \le Q$ with $\|d\xi\| \le 1/(Q+1) \le \sqrt{\varepsilon}$, and $\varepsilon d \le \sqrt{\varepsilon}$. Choose a shortest lattice vector v, so $\|v\|_\infty = s_1$. It is primitive and extends to a lattice basis (v, w). After subtracting an integer multiple of v, the parallel component of w has length at most $\|v\|_2/2$; its perpendicular component has length $\varepsilon/\|v\|_2$. Therefore

$$\|w\|_\infty \le \|w\|_2 \le s_1/\sqrt{2} + \varepsilon/s_1 < 2\varepsilon/s_1,$$

where $s_1^2 \le \varepsilon$ was used. The independent pair (v, w) proves the upper bound. □

The exponent identities. Write $\delta(q) = \|q\xi\|$ for irrational ξ and put $\omega = \mu(\xi) - 1$. Equivalently,

$$\omega = \limsup_{q \to \infty} \frac{-\log \delta(q)}{\log q}.$$

The first minimum satisfies

$$s_1(\varepsilon) = \min\left(1, \min_{q \ge 1} \max\{\delta(q), \varepsilon q\}\right).$$

If $\omega < \infty$ and $\eta > \omega$, then $\delta(q) \ge q^{-\eta}$ for all sufficiently large q. The minimum of $\max(q^{-\eta}, \varepsilon q)$ over positive real q is $\varepsilon^{\eta/(\eta+1)}$. The finitely many remaining q have fixed nonzero errors. Thus

$$\limsup_{\varepsilon \downarrow 0} \frac{-\log s_1(\varepsilon)}{-\log \varepsilon} \le \frac{\omega}{1+\omega}.$$

For every $0 < \eta < \omega$, infinitely many q satisfy $\delta(q) < q^{-\eta}$. At $\varepsilon_q = \delta(q)/q$, $s_1(\varepsilon_q) \le \delta(q)$, and

$$\frac{-\log s_1(\varepsilon_q)}{-\log \varepsilon_q} \ge \frac{-\log \delta(q)}{\log q - \log \delta(q)} > \frac{\eta}{1+\eta}.$$

Letting $\eta \uparrow \omega$ gives equality. If $\omega = \infty$, the same lower argument tends to one; the reverse inequality follows from $s_1(\varepsilon) \ge \varepsilon$ for $0 < \varepsilon \le 1$. Thus the first identity holds also for $\mu(\xi) = \infty$.

These limsup statements remain true on $\varepsilon_a = C_{2a}^{-1}$. Indeed

$$\frac{C_{2a+2}}{C_{2a}} = 4 - \frac{2}{a+1} \in [3, 4),$$

so any sufficiently small ε lies between two consecutive ε_a with ratio at most four. Multiplying ε by a factor in $[1/4, 4]$ changes S_ε, and hence its first minimum, by at most the same factor. The resulting logarithmic errors are bounded. Lemma 12.2 gives

$$\frac{-\log s_2(\varepsilon_a)}{\log C_{2a}} = 1 - \frac{-\log s_1(\varepsilon_a)}{\log C_{2a}} + o(1).$$

Finally (12.2) transfers both minima to $\lambda_i(2a)$ with logarithmic error $O(\log a)$, whereas $\log C_{2a} = a \log 4 + O(\log a)$. This proves (12.3)–(12.4).

If $\xi = P/Q$, the primitive zero direction is $(-P, Q)$. For sufficiently small ε, one has $s_1 = \varepsilon Q$ and $s_2 = 1/Q$: every vector off that line has error at least $1/Q$, and a Bezout pair with $Qb+Pd = 1$ realizes that error. Polynomial equivalence proves the stated logarithmic limits. In the moment norm itself, two independent integer pairs cannot both have zero actual output, so (12.1) gives $\lambda_2 \geq 1/Q$. □

Arithmetic meaning. The theorem separates an experimentally small second minimum from a proved arithmetic exponent. If the irrationality of V were established, its second-minimum lower decay exponent would determine $\mu(V)$ exactly. In particular a rigorously proved value strictly below $1/2$ would imply $\mu(V) > 2$ and, by Roth's theorem [3], transcendence. The present paper proves neither that inequality nor irrationality. The quantified lattice is an exact reformulation of this additional Diophantine difficulty, not its resolution.

13 Algebraic root exclusion and the missing transcendence inequality

The prime-scale determinant constructions have an arithmetic consequence stronger than nonvanishing at a fixed rational argument. Their coefficient certificates exclude all algebraic arguments of bounded absolute height, uniformly in the degree. This does not by itself make their value at the Gieseking constant small.

For a nonzero polynomial $P = \sum_{j=0}^{s} c_j X^j \in \mathbb{Z}[X]$, write $H(P) = \max_j |c_j|$. For an algebraic number ξ of degree D, with primitive integral minimal polynomial $A \prod_{i=1}^{D} (X - \xi_i)$, $A > 0$, use the absolute logarithmic height

$$h(\xi) = \frac{1}{D} \left(\log A + \sum_{i=1}^{D} \log \max(1, |\xi_i|) \right).$$

Every extension of the p-adic valuation below is normalized by $v_p(p) = 1$.

Theorem 13.1 (A uniform algebraic root-height certificate)**.** *Let p be a prime and $r \geq 1$ an integer. Suppose $P = \sum_{j=0}^{s} c_j X^j \in \mathbb{Z}[X]$, $s \geq 1$, satisfies*

$$v_p(c_0) = 0, \qquad v_p(c_j) \geq rj \quad (1 \leq j \leq s), \qquad v_p(c_s) = rs. \tag{13.1}$$

Then every root of P in $\overline{\mathbb{Q}}_p$ has valuation exactly $-r$. If ξ is any algebraic root of P and $D = [\mathbb{Q}(\xi) : \mathbb{Q}]$, the leading and constant coefficients of its primitive minimal polynomial satisfy

$$v_p(A) = rD, \qquad v_p(a_0) = 0.$$

In particular,

$$h(\xi) \geq r \log p. \tag{13.2}$$

Thus $P(\xi) \neq 0$ simultaneously for every algebraic ξ with $h(\xi) < r \log p$, with no restriction on its degree.

Proof. Let z be a p-adic root and write $t = v_p(z)$. The constant coefficient is nonzero, so $z \neq 0$. If $t > -r$, the constant summand in $P(z)$ has valuation zero, while every other summand has positive valuation. It cannot cancel. If $t < -r$, the leading summand has valuation $s(r+t)$, and every summand of degree $j < s$ has strictly larger valuation, since $v_p(c_j z^j) \geq j(r+t) > s(r+t)$. This is impossible as well. Therefore $t = -r$.

Let $M(X) = A\prod_{i=1}^{D}(X - \xi_i) = \sum a_k X^k$ be the primitive minimal polynomial of an algebraic root. Since M divides P over $\mathbb{Q}$, every p-adic conjugate ξ_i is a root of P and has valuation $-r$. The constant coefficient has exact valuation $v_p(a_0) = v_p(A) - rD$, while every elementary symmetric expression gives

$$v_p(a_k) \geq v_p(A) - r(D-k) = v_p(a_0) + rk.$$

Integrality gives $v_p(a_0) \geq 0$. If it were positive, every coefficient would be divisible by p, contrary to primitivity. Hence $v_p(a_0) = 0$ and $v_p(A) = rD$. Since $A \geq p^{rD}$, the defining height formula proves (13.2). □

Corollary 13.2 (Application to both prime-scale constructions)**.** *The primitive polynomial G_p of Heisei R40 [17], of degree $(p-1)/2$, and the primitive polynomial $G_{a,p}$ of Meiji R57, of degree p at a prime good for its fixed positive shift a, satisfy*

$$G_p(\xi) \neq 0, \qquad G_{a,p}(\xi) \neq 0 \quad \textit{whenever} \quad h(\xi) < 2\log p.$$

In each case every algebraic root has height at least $2\log p$. In particular, for every fixed algebraic ξ, either family is nonzero at ξ for every sufficiently large eligible prime.

Proof. Both source constructions prove $v_p(c_0) = 0$, $v_p(c_j) \geq 2j$, and equality $v_p(c_s) = 2s$ at the leading coefficient. Apply Theorem 13.1 with $r = 2$. The good-prime restriction in Meiji R57 excludes only finitely many primes for each fixed positive shift. The shift is fixed before p varies. □

This is a consequence of the already established coefficient certificates, not a new proof of those determinant constructions or a claim of a new general height principle. It closes their algebraic-specialization nonvanishing question, including all degrees and heights below the stated threshold. An arbitrary rational rescaling before primitive normalization does not preserve the coefficient hypothesis and must not be substituted.

Lemma 13.3 (A norm lower bound with the degree explicit)**.** *Let ξ be algebraic of degree D and let $0 \neq P \in \mathbb{Z}[X]$ have degree s, with $P(\xi) \neq 0$. Then*

$$\log|P(\xi)| \geq -(D-1)\log H(P) - (D-1)\log(s+1) - Ds\,h(\xi). \tag{13.3}$$

Proof. Set $\xi_1 = \xi$ and let M be the primitive minimal polynomial above. Its nonzero resultant with P is an integer, so

$$1 \leq |\operatorname{Res}(M,P)| = A^s \prod_{i=1}^{D} |P(\xi_i)|.$$

For $i \geq 2$, $|P(\xi_i)| \leq (s+1)H(P)\max(1,|\xi_i|)^s$. Also $A\prod_{i=2}^{D}\max(1,|\xi_i|) \leq \exp(Dh(\xi))$. Isolating the first factor and taking logarithms gives the assertion. □

Theorem 13.4 (A degree-exclusion and transcendence criterion)**.** *Let x be a real number. For infinitely many primes $p \to \infty$, let $P_p \in \mathbb{Z}[X]$ be nonzero polynomials which satisfy (13.1) for some fixed $r \geq 1$. Suppose there is a positive scale $T_p \to \infty$ such that*

$$\deg P_p = o(T_p), \qquad \log H(P_p) \leq BT_p + o(T_p), \qquad \log|P_p(x)| \leq -AT_p + o(T_p),$$

where $A > 0$, $B \geq 0$, and $\log 0 = -\infty$. Then x cannot be algebraic of a degree D for which

$$(D-1)B < A. \tag{13.4}$$

In particular, $B = 0$ implies that x is transcendental.

Proof. If x were algebraic, Theorem 13.1 would make $P_p(x)$ nonzero for all sufficiently large primes in the sequence. Apply Lemma 13.3. Its last term is $o(T_p)$ because x and D are fixed, and its logarithmic degree term is also $o(T_p)$. The result is $\log|P_p(x)| \geq -(D-1)BT_p - o(T_p)$, contradicting the assumed upper bound whenever (13.4) holds. □

For the two constructions in Corollary 13.2, the degrees are $O(p)$, so $T_p = p^2$ is permissible. A hypothetical bound $\log|P_p(V)| \leq -Ap^2 + o(p^2)$ would prove irrationality without a separate height bound: take $D = 1$ in (13.3). To exclude quadratic values as well one needs, for example, $\log H(P_p) \leq Bp^2 + o(p^2)$ with $A > B$. To establish transcendence by this route, it would suffice to prove the same strict quadratic decay together with $\log H(P_p) = o(p^2)$. None of these required decay estimates is established for the present primitive determinant families. Their known local nonvanishing and formal normality do not provide the missing inequality at the real place.

The external arithmetic result and the field of scalars. Calegari, Dimitrov and Tang prove the $\mathbb{Q}$-linear independence of $1, \pi^2, L(2, \chi_{-3})$ [1]. Their later effective result gives the bound $|L(2, \chi_{-3}) - u/v| > v^{-24781}$ outside a finite computable set of reduced rational approximants [2]. Neither statement supplies $\mathbb{Q}(\sqrt{3})$-linear independence. In particular, if V were a nonzero rational number, then $L(2, \chi_{-3}) = 4V/(3\sqrt{3})$ would be a quadratic irrational, consistent with both statements. Thus the classical irrationality of the scaled constant cannot replace the real-place estimate in Theorem 13.4.

14 What is closed, and what remains

The factor-three inverse theorem closes an explicit denominator question. The first-layer theorem closes an image classification in its no-wrap range and, with the nested-image input, a missing prime interval. The dyadic index determines an intrinsic arithmetic cost which cannot be removed by changing the rational output basis. The positive-family optimization is a global theorem for its stated budget, and the calibrated lattice theorem identifies its exact Diophantine content.

These results also prevent several invalid shortcuts. Rational surjectivity does not produce integral controls of comparable size. A positive rational kernel does not bound the height of an integral representative. A nonzero formal coordinate vector need not give a nonzero real form. A small unnormalized determinant can become large after primitive normalization. A bound for one successive minimum cannot replace a bound for several independent directions.

Two concrete infinite problems remain sufficient for an arithmetic conclusion. One is the existence of the positive, nuisance-cancelling, integral-output controls of Corollary 7.3 within its proved height budget. The other is strict quadratic real-place decay of a primitive prime-scale polynomial, with the coefficient height needed for the desired degree exclusion in Theorem 13.4. These are genuine additional estimates, not consequences of the local algebra proved here.

There is a useful distinction between the two basis results. The inverse denominator theorem concerns rational coefficients in the complete $4n$ window and improves their arithmetic cost. The dyadic obstruction concerns a different odd space and a complete collection of integer-output controls. They cannot be identified in either direction. Similarly the new optimized parameter does not inherit a positive-cone theorem proved at a different parameter without another argument.

A Source coverage and reproducibility

The supplied corpus contains the following Gieseking manuscripts. The seven Mills/Pisot notes in the additional archive are outside the present synthesis.

Series	Earlier notes	Additional notes	Total
Keio	22	8	30
Reiwa	20	12	32
Meiji	37	9	46
Taisho	24	6	30
Heisei	33	12	45
Total	136	47	183

The companion verification archive includes a title/path/hash catalogue, five series audits, the arithmetic audit, and exact checkers. The audits state which proof mechanisms were examined closely and which older results were superseded. This is not a claim that every historical calculation or every source-mentioned external checker was independently rerun.

The finite replays include: the complete mixed-moment determinant through $n = 5$; first and second prime-degree thresholds and simple/double pole witnesses for 82 pairs (p, m); 95 complete saturation-source spaces; the dyadic index through $n = 25$ and the full odd output identities through $n = 10$; conic coordinate denominator and normalizer checks; and exact rational enclosures for the optimization constants. The new normalized inverse estimates are also checked independently on finite moment matrices. No finite replay is used as a substitute for an all-index proof.

For the new theorems, separate proof reviews covered the factor-three carry argument, the dyadic Smith index, the saturation interval, the positive budget optimization, and the calibrated minima. These are internal checks, not external peer review or a certification of historical novelty. The published results cited below are used only with their stated scalar fields and hypotheses.

References

[1] F. Calegari, V. Dimitrov and Y. Tang, *The linear independence of* 1, $\zeta(2)$, *and* $L(2, \chi_{-3})$, arXiv:2408.15403v2 (2024). https://arxiv.org/abs/2408.15403.

[2] F. Calegari, V. Dimitrov and Y. Tang, *Arithmetic holonomy bounds and effective Diophantine approximation*, arXiv:2510.04156 (2025), 2026 SIAM proceedings version; Theorem 5.1. https://arxiv.org/abs/2510.04156.

[3] K. F. Roth, *Rational approximations to algebraic numbers*, Mathematika **2** (1955), 1–20. https://doi.org/10.1112/S0025579300000644.

[4] Keio, Reiwa, Meiji, Taisho and Heisei series, *Supplied Gieseking research corpus*: 136 earlier notes and 47 subsequent notes (2026). Unpublished manuscripts supplied for this synthesis; individual identifiers below are corpus labels, not journal citations.

[5] Taisho R05, *A Five-Constant Moment Algebra for the Gieseking Volume.*

[6] Taisho R13, Frobenius moment descent, supplied manuscript.

[7] Taisho R28, low-degree local images and boundary model, supplied manuscript.

[8] Taisho R36, *Arithmetic and Decay Bounds for Proportional Bernstein–Rodrigues Controls.*

[9] Taisho R37, *Uniform Positive Bernstein Cones and a Three-Dimensional Weighted Lattice Reduction.*

[10] Taisho R38, *Sharp Prime-Degree Thresholds for Mixed Gieseking Period Images.*

[11] Meiji R58, *An Exact Four-Block Determinant and Minimal-Window Gieseking Moment Normality.*

[12] Meiji R59, *A Calibrated Two-Dimensional Moment Lattice.*

[13] Meiji R61, *An Explicit Full-Window Moment Inverse with Exponential-Height Rational Coefficients.*

[14] Meiji R62, *Laurent and Centered-Angle Primitives for the Full Gieseking Moment Inverse.*

[15] Meiji R57, *Positive Block Shifts Remove a Prime-Scale Gieseking Determinant Rank Obstruction.*

[16] Heisei R06, *An Exact Determinant and All-Size Normality for Gieseking Moment Cancellation.*

[17] Heisei R40, primitive prime-scale determinant construction, supplied manuscript.

[18] Heisei R42, continuous-row/common-positive-weight odd-column sign regularity, supplied manuscript.

[19] Heisei R43, *An Explicit Binomial Adjoint Factorization and Small Denominators in Tail Gieseking Rows.*

[20] Heisei R48v2, direct odd rational-output construction and coefficient denominators, supplied manuscript.

[21] Heisei R49, *The Sharp Large-Prime Residue Rank of the Odd Rational-Output Basis.*

[22] Heisei R50, *The Exact Asymptotic Denominator of the Direct Odd Output Basis.*

[23] Keio R49, *Normality of Three Consecutive Legendre Density Periods.*

[24] Keio R50, *Four-Consecutive Conic Normality and All-Index High-Block Normalizers.*

[25] Keio R51, *A New Endpoint Source and a Two-Block Local Image.*

[26] Reiwa R49, complete coefficient-saturation basis and Frobenius descent, supplied manuscript.

[27] Reiwa R52, two-mode saturation bands and the remaining two-block interval, supplied manuscript.